

EDIT WHO SPEAKS, CONTROL HOW THEY SPEAK: GLOBAL TIMBRE EDITING AND LOCAL INSTRUCTION CONTROL FOR TTS

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ABSTRACT

Instruction-based text-to-speech (TTS) offers control over voice characteristics and speech expression through interfaces including voice cloning and text-based voice design. Voice cloning reproduces a reference voice, whereas text-based voice design creates a voice from a natural-language description. However, neither interface directly enables users to modify the timbre of a given reference and synthesize speech with the modified voice. Meanwhile, utterance-level expressive instructions leave changes across individual text segments underspecified. We introduce **EDICT**, a framework that unifies global timbre editing and local expressive control by using an edited acoustic reference to anchor voice identity across segments. To enable synthesis with an instruction-edited voice, EDICT combines reference audio with structured timbre edits to generate an edited reference in codec-token space. This representation serves as a shared voice anchor for a frozen TTS backbone, allowing segment-specific natural-language instructions to guide expression. To accommodate instruction changes while supporting acoustic continuity, EDICT rebuilds the KV cache at each segment boundary, refreshing instruction conditioning while retaining bounded acoustic context from previously generated speech. Evaluations on our proposed TimbreEdit-Bench and IntraTTS-Bench demonstrate improved timbre editing and a favorable balance between local instruction adherence, speaker consistency, and transition quality. Audio demos are available¹.

1 INTRODUCTION

Instruction-based TTS not only synthesizes speech from the given text but also allows users to explicitly control voice characteristics and speech delivery (Guo et al., 2023; Yang et al., 2024; Zhang et al., 2026). Narration and dialog often require control at two temporal scopes: voice characteristics should remain consistent throughout an utterance, while speech delivery may vary across individual text segments. For example, a brighter version of a reference voice may serve as a shared voice anchor throughout an utterance, while the delivery changes from calm to excited and finally to soft. We study TTS requests that combine *global* editing of a reference voice with *local* instructions for successive segments of one utterance.

At the global level, existing approaches specify a target voice either through an acoustic reference, as in voice cloning, or through a textual description, as in voice design (Wang et al., 2023; Leng et al., 2024; Hu et al., 2026a; Lee et al., 2026; Zhao et al., 2025). Reference-conditioned editing provides a complementary interface: users specify changes relative to an existing voice, while unspecified attributes are intended to remain unchanged. Based on text-guided voice conversion and editing (Hai et al., 2024; Anastassiou et al., 2024b; Huang et al., 2024), we use the resulting edited voice

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¹<https://danny-nus.github.io/EDICT-demopage/>

as a shared conditioning reference for synthesizing new text under changing delivery instructions, thereby integrating timbre editing into the TTS pipeline.

At the local level, segment-specific instructions must align with their text spans while preserving the globally edited voice characteristics and maintaining natural transitions. Most instruction-based TTS systems use a single prompt for the entire utterance, while only a few support the association of individual text spans with their own instructions within an utterance. TED-TTS establishes segment-aware emotion and duration control, but its masking mechanism does not refresh the acoustic history, allowing cached key-value (KV) states to carry preceding conditions into later segments and cause *control leakage* (Wu et al., 2024; Jeong et al., 2025; Lemerle et al., 2025; Liang et al., 2026). Retaining the full history may weaken new instructions, whereas discarding it removes the context needed for natural continuation. Kang et al. mitigate *style self-referencing* through KV-cache swapping and sliding-window attention, but focus on a single source-to-target transition and report a trade-off between transition strength and speaker similarity (Kang et al., 2026). EDICT rebuilds the cache at every boundary from the current instruction, edited reference, and bounded acoustic context, enabling repeated instruction switching while preserving identity and natural transitions.

We introduce **EDICT**, a controllable TTS framework whose trainable timbre editor maps a global request to attribute operations and edits the source voice, producing a codec reference shared across all segments of a frozen TTS backbone. To mitigate control leakage, EDICT reconstructs the KV cache at instruction boundaries by recomputing selected initial and recent acoustic states conditioned on the next text, instruction, and shared reference. Initial context provides voice cues, while recent context supports continuity. We propose TimbreEdit-Bench and IntraTTS-Bench to evaluate, respectively, requested voice edits and preservation of unspecified attributes, and local instruction following with consistent speaker identity and natural transitions. Experiments on Qwen3-TTS-VoiceDesign (Qwen3-TTS-VD) and CosyVoice 2 assess KV cache reconstruction across backbones and its combination with timbre editing, showing improved instruction following over the respective baselines for timbre editing and intra-utterance control.

In summary, this paper makes the following contributions:

- **Reference-conditioned timbre editing for TTS.** We develop a timbre editor that modifies a source voice through relative instructions, producing a reusable codec reference for new texts and delivery conditions.
- **Segment-level control through KV cache reconstruction.** We reconstruct the retained acoustic context under new instructions to mitigate control leakage and balance local adherence with continuity, without additional backbone training.
- **Bilingual benchmarks and compositional evaluation.** We introduce TimbreEdit-Bench and IntraTTS-Bench to evaluate global timbre editing, local instruction control, and their composition through objective metrics, listening tests, and ablations.

2 RELATED WORK

Language-based voice and style control. Early work represented speaking style through learned latent tokens (Wang et al., 2018). PromptTTS, InstructTTS, PromptStyle, and TextrolSpeech control voice and delivery through descriptions (Guo et al., 2023; Yang et al., 2024; Liu et al., 2023; Ji et al., 2024). More recent systems use language models to condition speech generation on voice and delivery instructions (Zhou et al., 2024; Du et al., 2025; Hu et al., 2026a; Yang et al., 2025b; Zhang et al., 2026). ControlSpeech and FlexiVoice combine voice references with language-based style control; ReStyle-TTS adjusts style relative to a reference, and Vevo separates timbre and style prompts (Ji et al., 2025; Chen et al., 2026; Li et al., 2026a; Zhang et al., 2025).

Reference-anchored voice editing. Voice cloning specifies voices acoustically (Wang et al., 2023; Le et al., 2023; Du et al., 2024), while voice design specifies them textually (Leng et al., 2024; Shimizu et al., 2024; Hu et al., 2026b). Voice editing applies instructed changes to a source voice (Huang et al., 2024). VoxEditor represents relative attribute edits (Sheng et al., 2025); DreamVoice and TES-VC use text-guided conversion (Hai et al., 2024; Jin et al., 2025); VoiceShop supports selective voice changes (Anastassiou et al., 2024b). Recent systems integrate editing with TTS or unify generation and editing (Pei et al., 2026; Hai et al., 2026; Wang et al., 2026a). EDICT

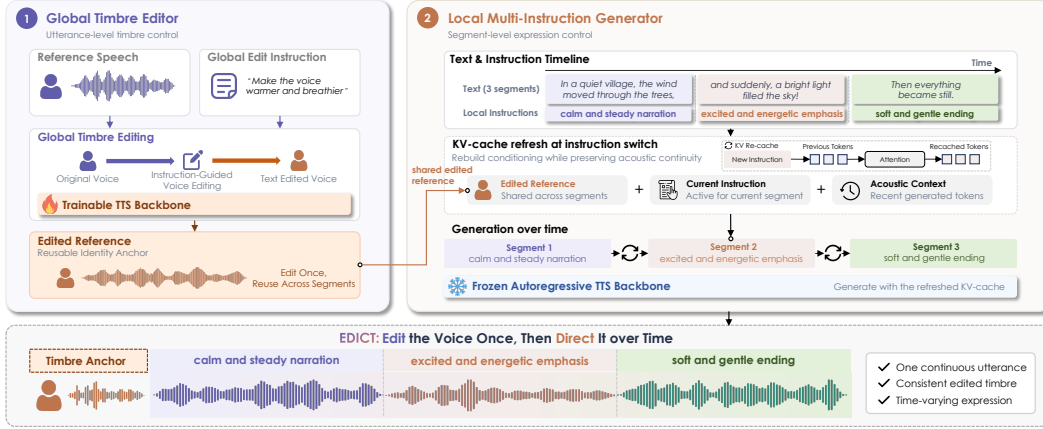


Figure 1: **EDICT combines global voice editing with local delivery control.** The trainable editor uses the source voice and global edit instruction to generate an edited codec reference. This reference conditions every segment of the frozen TTS backbone, whose KV cache is rebuilt at instruction boundaries using selected acoustic context. The segments form one continuous utterance.

produces a reusable edited codec reference for subsequent TTS under changing local instructions. We evaluate requested changes and preservation separately, since target-voice matching does not establish preservation of unspecified attributes.

Local control and acoustic history. Time-varying style and local prosody (Chen & Rudnicky, 2022; Lenglet et al., 2023; Wu et al., 2024; Jeong et al., 2025) have expanded to word-level styles, emotion trajectories, and timing control (Wang et al., 2025a; Liu et al., 2026; Zhou et al., 2026b; Mai et al., 2026; Lemerle et al., 2025). For training-free control, TED-TTS uses segment-aware causal masking and monotonic alignment (Liang et al., 2026). Kang et al. address acoustic *style self-referencing* through KV-cache swapping and sliding-window attention for a source-to-target transition (Kang et al., 2026). EDICT recomputes retained acoustic states at successive instruction switches and uses an edited reference throughout the utterance. Appendix A defines the tasks and comparison scope.

3 METHODOLOGY

3.1 TASK FORMULATION AND OVERVIEW

At inference, an LLM parses the global request into attribute operations, the timbre editor produces an edited codec reference shared across segments, and a frozen TTS backbone generates the segments sequentially while rebuilding its KV cache at local-instruction boundaries. Specifically, given a reference codec sequence $\mathbf{c}_{\text{ref}}$, a timbre-edit request $\mathcal{I}_{\text{edit}}$, and N text-instruction pairs $\{(\mathbf{x}_k, \mathcal{I}_k)\}_{k=1}^N$, EDICT generates successive segments in a shared edited voice. As shown in Figure 1, a learned editor p_θ and a frozen TTS model p_ϕ form a two-stage cascade:

$$\mathbf{c}'_{\text{ref}} \sim p_\theta(\cdot \mid \mathbf{c}_{\text{ref}}, \mathbf{z}_{\text{edit}}, \mathbf{x}_{\text{edit}}), \quad \mathbf{c}_{1:N} \sim p_\phi(\cdot \mid \mathbf{c}'_{\text{ref}}, \{(\mathbf{x}_k, \mathcal{I}_k)\}_{k=1}^N), \quad (1)$$

where $\mathbf{z}_{\text{edit}}$ encodes the requested attribute operations and $\mathbf{x}_{\text{edit}}$ specifies the text spoken in the edited reference. The edited codec reference conditions every segment, while local instructions change their delivery. Segment-boundary reconstruction produces one continuous codec stream for waveform decoding.

3.2 REFERENCE-CONDITIONED TIMBRE EDITING

Attribute-based instruction encoding. Following work on relative voice editing and perceptual voice-quality control (Sheng et al., 2025; Rautenberg et al., 2025), we encode edits along nine attributes: perceived gender, age impression, pitch register, brightness, vocal weight, resonance focus,

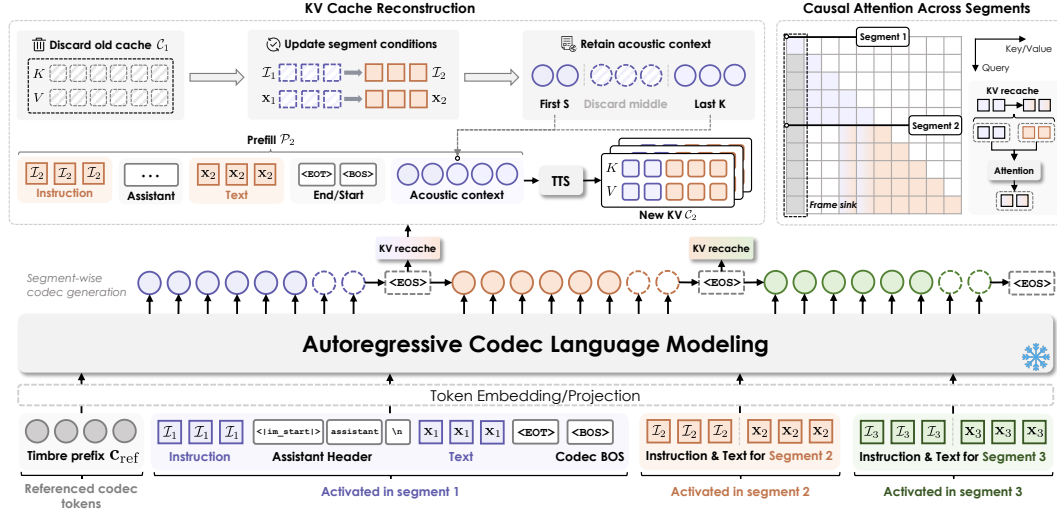


Figure 2: **Segment-boundary KV cache reconstruction.** Each non-final $\langle \text{EOS} \rangle$ triggers a fresh prefill containing the next instruction and text, the shared edited reference, and acoustic inputs from the first S and most recent K generated positions. The upper-left panel shows replacement of the old cache; the upper-right panel shows causal attention after reconstruction. The lower timeline illustrates successive instruction switches within one codec stream.

roughness, breathiness, and nasality. Gender and age are categorical; the others use three ordered levels, with definitions and values in Table 9 (Appendix B). At inference, an LLM maps structured or descriptive $\mathcal{I}_{\text{edit}}$ to operations specifying direction and magnitude for ordinal changes, target categories for categorical changes, and $\langle \text{keep} \rangle$ for unmentioned attributes. For example, $\langle \text{brightness_up_1} \rangle$ denotes low-to-mid or mid-to-high, while $\langle \text{brightness_up_2} \rangle$ denotes low-to-high. Operations are serialized in a fixed attribute order into $\mathbf{z}_{\text{edit}}$ and interpreted together with the acoustic reference. During training, profile differences determine the operations.

Paired supervision. Qwen3-14B (Yang et al., 2025a) describes source and target profiles, and Qwen3-TTS-VD (Hu et al., 2026a) synthesizes their Chinese and English speech under a shared neutral-delivery requirement. The requested edits and keep operations are derived from differences between the source and target profiles. Training tuples $(\mathbf{c}_{\text{ref}}, \mathbf{z}_{\text{edit}}, \mathbf{x}_{\text{edit}}, \mathbf{c}_{\text{tgt}})$ cover single-attribute and compositional edits with same or different texts, within and across languages. The target transcript specifies the edited reference’s content, allowing it to differ from the source. Appendix B details construction and filtering.

Training. We initialize the editor from Qwen3-TTS-Base (Hu et al., 2026a) and perform supervised fine-tuning (SFT) to predict target codecs conditioned on $g = (\mathbf{c}_{\text{ref}}, \mathbf{z}_{\text{edit}}, \mathbf{x}_{\text{edit}})$, using teacher-forced cross-entropy over all codebooks at target audio positions. We then apply reward-weighted DPO (Rafailov et al., 2023) with auxiliary SFT, denoted wDPO. Candidates sampled from the frozen SFT model p_{ref} are scored by WavLM speaker-embedding cosine similarity to the paired target (Chen et al., 2022). These scores determine preference pairs $(\mathbf{c}_g^+, \mathbf{c}_{g,j}^-)$ and positive reward-gap weights $w_{g,j}$. Defining $q_\theta(\mathbf{c} | g) = \log[p_\theta(\mathbf{c} | g)/p_{\text{ref}}(\mathbf{c} | g)]$, we minimize:

$$\mathcal{L}_{\text{wDPO}} = - \frac{\sum_{(g,j) \in \mathcal{B}} w_{g,j} \log \sigma(\beta [q_\theta(\mathbf{c}_g^+ | g) - q_\theta(\mathbf{c}_{g,j}^- | g)])}{\sum_{(g,j) \in \mathcal{B}} w_{g,j}} + \lambda_{\text{SFT}} \mathcal{L}_{\text{SFT}}, \quad (2)$$

where $\mathcal{B}$ is a minibatch of preference pairs, σ is the sigmoid, β scales the preference logit, and $\mathcal{L}_{\text{SFT}}$ is the codec cross-entropy on chosen candidates. Preference pairs and weights remain fixed during post-training. Appendix D.1 details candidate sampling, pair selection, and training settings.

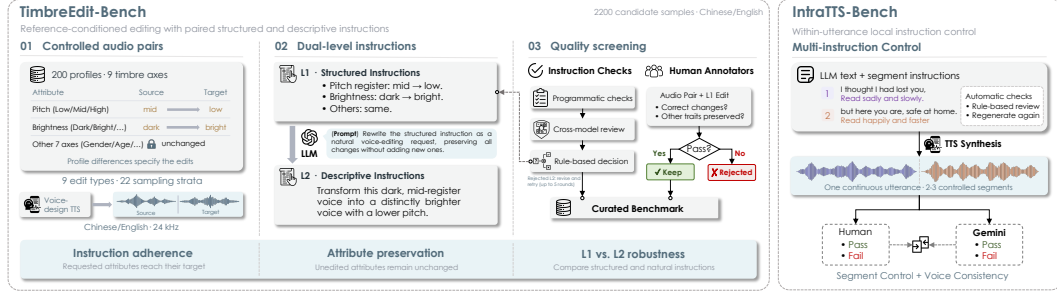


Figure 3: **Benchmark construction and evaluation.** Left: TimbreEdit-Bench pairs source and target voices with editing instructions. Right: IntraTTS-Bench aligns delivery instructions with successive text segments.

3.3 SEGMENT-LEVEL INSTRUCTION CONTROL

The edited reference $\mathbf{c}'_{\text{ref}}$ remains fixed across segments, while $\mathcal{I}_k$ controls each segment’s emotion, speaking rate, or prosody. This stage uses a frozen autoregressive TTS backbone and requires no additional training. Figure 2 shows how generation proceeds across instruction boundaries. Segment-aware masks such as those in TED-TTS (Liang et al., 2026) restrict direct access to other segments’ conditions, but historical acoustic states may still encode those conditions. Drawing on LongLive’s recaching strategy (Yang et al., 2025c), we reconstruct these states under the next instruction.

KV cache reconstruction. A non-final $\langle \text{EOS} \rangle$ triggers replacement of $(\mathcal{I}_k, \mathbf{x}_k)$ with $(\mathcal{I}_{k+1}, \mathbf{x}_{k+1})$. We assemble a fresh prefill $\mathcal{P}_{k+1}$ containing the new instruction, shared edited reference, new text, and native template and boundary tokens. We then append acoustic input embeddings for the first S and most recent K generated codec positions. The old KV cache is discarded, and a causal forward pass through the frozen model reconstructs the entire prefill cache:

$$\mathcal{C}_{k+1} = \text{Prefill}_{\phi}(\mathcal{P}_{k+1}) = \left\{ \left(\mathbf{K}_{\mathcal{P}_{k+1}}^{(\ell)}, \mathbf{V}_{\mathcal{P}_{k+1}}^{(\ell)} \right) \right\}_{\ell=1}^L, \quad (3)$$

where L is the number of Transformer layers. Acoustic input embeddings are reused, but their keys and values are recomputed under the updated conditions. Initial context retains evidence from the utterance’s start; recent context supplies immediate continuation history. Reconstruction bounds the retained history, while the cache grows normally within each segment. Section 5.5 tests the role of each context source and compares reconstruction with reuse of the old KV states.

Continuous decoding. For a single attention head, let j index an acoustic input position processed after the reconstructed prefill. The query $\mathbf{q}_{k+1,j}$ attends to the rebuilt prefill and the current segment’s acoustic inputs through position j (layer indices omitted):

$$\text{Attn}(\mathbf{q}_{k+1,j}) = \text{softmax} \left(\frac{\mathbf{q}_{k+1,j} [\mathbf{K}_{\mathcal{P}_{k+1}}; \mathbf{K}_{k+1,\leq j}]^T}{\sqrt{d_h}} \right) [\mathbf{V}_{\mathcal{P}_{k+1}}; \mathbf{V}_{k+1,\leq j}], \quad (4)$$

where $[\cdot]$ denotes concatenation along the sequence dimension, d_h is the head dimension, and $(\mathbf{K}_{k+1,\leq j}, \mathbf{V}_{k+1,\leq j})$ contains the current segment’s KV states through input position j . These positions are already observed inputs; the attention output is used to predict the next codec frame. History omitted from the cache remains in the output codec stream. Non-final $\langle \text{EOS} \rangle$ markers trigger reconstruction without entering the acoustic output; the final $\langle \text{EOS} \rangle$ ends generation. All segments are decoded into one waveform. Appendix D.2 details decoding, and Appendix G visualizes attention patterns.

4 EXPERIMENTS

4.1 DATA AND BENCHMARKS

Training data. We construct 1M directed editing pairs from a synthetic Chinese–English speech bank generated by Qwen3-TTS-VD using nine-attribute timbre profiles. We use 5,800 profiles for

training and hold out 200, excluding held-out profiles from both sides of every training pair. Edit operations are derived from source–target profile differences, covering single-attribute and compositional edits within and across languages. Offline post-training conditions are sampled from the same training pairs. Appendix B details the construction and data distributions.

Evaluation benchmarks. **TimbreEdit-Bench** contains 1,955 source–target pairs from 200 held-out profiles across nine edit categories. Three annotators screen for audible edits, attribute preservation, and audio quality; retained pairs have semantically matched structured and descriptive instructions. **IntraTTS-Bench** contains 997 requests with two or three text segments and corresponding delivery instructions to evaluate local control and continuity. Both benchmarks cover Chinese and English. Figure 3 summarizes the workflow, and Appendix C details construction and screening.

4.2 IMPLEMENTATION DETAILS

We initialize the editor from Qwen3-TTS-12Hz-1.7B-Base and fine-tune all parameters, including codec predictors. SFT is followed by one round of wDPO using preference pairs constructed from 20K training conditions, with $M = 16$ sampled candidates per condition. We set the DPO scale $\beta = 0.1$ and auxiliary SFT coefficient $\lambda_{\text{SFT}} = 0.01$. Qwen3-14B parses editing requests. Local generation uses frozen Qwen3-TTS-VD, with CosyVoice 2 as a second backbone and default context $(S, K) = (4, 5)$. For local control, **Concat.** synthesizes segments independently, **Joint** prompts the complete utterance in one pass, and our **TED-TTS** adaptation switches segment conditions through attention masking and monotonic alignment without KV cache reconstruction. Appendix D provides the remaining training settings and baseline and ablation configurations.

4.3 EVALUATION METRICS

Chinese **CER** and English **WER** assess intelligibility; **UTMOS** (Saeki et al., 2022) and **DNS-MOS OVRL** (Reddy et al., 2022) estimate perceptual quality. WavLM (Chen et al., 2022) and ERes2Net (Chen et al., 2023) measure target/source similarity (**SIM-T/R**); WavLM also measures inter-segment similarity (**SIM-I**). Following InstructTTSEval (Huang et al., 2025), Gemini 3.1 Pro evaluates instruction compliance (**SegAcc**), requested emotion (**EmoAcc**), transitions (**Trans**), and attribute-level editing (**Up/Down**, **Category**, and **Preservation**). Human evaluation uses five-point ratings of naturalness (**N-MOS**), edit accuracy (**Edit-MOS**), target similarity (**S-MOS**), instruction following (**Instr-MOS**), and transitions (**Trans-MOS**), with 95% confidence intervals from audio-item bootstrapping. Appendix D provides training settings, baseline and ablation configurations.

5 RESULTS

5.1 TIMBRE EDITING

Table 1 shows that *EDICT-Stage I* (Section 3.2) achieves the highest SIM-T among instruction-conditioned systems under both instruction forms, with gains on both ERes2Net and WavLM, the training-reward encoder. Unlike these baselines, its outputs are closer to the target than the source, with competitive predicted quality but not consistently the lowest recognition error. FreeVC (Li et al., 2023) and Seed-VC receive target-speaker audio and are evaluated on a separate same-text subset, providing an upper-bound setting with directly available target timbre. Our setting instead infers target timbre from source audio and an editing instruction. Results with human-recorded ESD references appear in Appendix H.1. Since target similarity does not establish attribute preservation, Table 2 evaluates requested operations and unchanged attributes separately. *EDICT-Stage I* leads in edit direction and categorical accuracy under both instruction forms, while Qwen3-TTS-VD better preserves unchanged attributes but follows edits less reliably, highlighting a trade-off between edit accuracy and preservation. Up/Down measures direction rather than magnitude; Appendix H.2 details the protocol.

5.2 INTRA-UTTERANCE INSTRUCTION CONTROL

Table 3 exhibits a consistent trend across Qwen3-TTS-VD and CosyVoice 2: the system with individual synthesis and concatenation (*Concat.*) yields the strongest segment-level and emotion accu-

Table 1: **Objective results on TimbreEdit-Bench.** WER/CER reports English WER / Chinese CER in percent. SIM-T/R denotes target/source similarity; Δ = SIM-T – SIM-R. Best and second-best scores within each instruction form are **bold** and underlined, with CER and WER ranked separately; SIM-R is unranked. Target-reference voice conversion receives target audio and is evaluated separately on a same-text subset, with its best similarity and Δ scores in **bold**.

System	Instruction	Content and quality			WavLM			ERes2Net		
		WER/CER (%)↓	UTMOS↑	DNSMOS↑	SIM-T↑	SIM-R	Δ ↑	SIM-T↑	SIM-R	Δ ↑
Instruction-conditioned systems										
Qwen3-TTS-VD	Structured	4.5 / 0.3	3.861	3.946	0.701	0.871	−0.170	0.450	0.745	−0.295
	Descriptive	3.0 / 0.3	3.830	3.965	0.676	0.832	−0.156	0.435	0.714	−0.279
Qwen3-TTS-Base	Structured	5.1 / 1.9	3.742	3.805	0.632	0.950	−0.318	0.431	0.845	−0.414
	Descriptive	3.3 / 1.1	3.756	3.776	0.637	0.950	−0.313	0.434	0.849	−0.415
CosyVoice 2	Structured	6.9 / 4.2	3.941	4.065	0.646	0.918	−0.272	0.396	0.825	−0.429
	Descriptive	7.9 / 2.5	4.021	4.104	0.644	0.937	−0.293	0.419	0.854	−0.435
EDICT-Stage I	Structured	4.1 / 1.7	3.830	3.942	0.819	0.620	0.199	0.653	0.442	0.211
	Descriptive	4.0 / 1.1	3.816	3.963	0.793	0.598	0.195	0.648	0.441	0.207
Target-reference voice conversion										
FreeVC	N/A	3.2 / 2.0	3.911	3.968	0.873	0.671	0.202	0.570	0.387	0.183
Seed-VC	N/A	3.0 / 2.5	3.342	3.893	0.957	0.659	0.298	0.889	0.463	0.427

Table 2: **Attribute-operation accuracy (%) on 500 TimbreEdit-Bench requests.** Gemini compares source and generated audio without the target recording. Up/Down assesses ordinal direction, Category assesses categorical targets, and Preservation assesses unchanged attributes. Best and second-best scores within each instruction form are **bold** and underlined.

System	Instruction	Up Acc.↑	Down Acc.↑	Category Acc.↑	Preservation↑
Qwen3-TTS-VD	Structured	67.0	64.3	53.2	83.3
	Descriptive	60.0	63.0	49.8	85.0
Qwen3-TTS-Base	Structured	62.3	54.5	46.4	45.7
	Descriptive	66.0	54.5	50.0	45.6
CosyVoice 2	Structured	<u>68.1</u>	<u>65.5</u>	<u>57.7</u>	<u>68.2</u>
	Descriptive	<u>72.2</u>	<u>73.3</u>	<u>69.0</u>	66.2
EDICT-Stage I	Structured	70.8	80.6	89.3	66.7
	Descriptive	77.9	79.1	91.7	<u>70.1</u>

racy but weaker transitions, while *Joint* favors speaker consistency over local instruction adherence. *EDICT-Stage II* achieves the best transitions and consistently improves SegAcc, EmoAcc, and SIM-I over TED-TTS across both backbones. Despite the different absolute control scores of the two backbones, *EDICT-Stage II* consistently improves SegAcc, EmoAcc, and transition quality over TED-TTS. It indicates that the benefit of KV cache reconstruction extends beyond Qwen3-TTS-VD. This improvement comes with higher WER/CER than the *Joint* setting, but only modest increases over TED-TTS. Speaker consistency remains comparable to TED-TTS, while predicted speech quality changes only slightly. Overall, KV cache reconstruction improves local instruction adherence and transition quality with a modest trade-off in content fidelity. Appendix H.3 reports complementary MED-TTS results.

5.3 JOINT TIMBRE AND DELIVERY CONTROL

We combine the reference or edited audio from TimbreEdit-Bench with the texts and delivery instructions from IntraTTS-Bench. *Qwen3-TTS-VD (Joint)* and *Qwen3-TTS-VD (Concat.)* take the original reference audio and a global editing instruction as input. *EDICT Stage I + Concat.* and the EDICT pipeline use the same edited reference, followed by either independent synthesis and concatenation or continuous generation with KV cache reconstruction, respectively. Table 4 highlights the importance and complementary contributions of the two stages. Compared with *Qwen3-TTS-VD (Concat.)*, incorporating the timbre editor (*EDICT Stage I*) improves target matching, but independently synthesized segments and concatenation (*Concat.*) still exhibit weak transitions in *EDICT Stage I + Concat.*. By further replacing the *Concat.* with KV cache reconstruction, EDICT

Table 3: **Intra-utterance instruction control on IntraTTS-Bench.** SegAcc, EmoAcc, and Trans measure instruction adherence, emotion realization, and transition quality in percent. SIM-I measures inter-segment speaker similarity; WER/CER reports English WER / Chinese CER in percent. Best and second-best scores within each backbone are **bold** and underlined, with CER and WER ranked separately.

System	Control (%)			Consistency	Content and quality		
	SegAcc↑	EmoAcc↑	Trans↑		WER/CER (%)↓	UTMOS↑	DNSMOS↑
Qwen3-TTS-VD (Concat.)	92.44	92.66	60.68	0.658	<u>4.5</u> / 0.9	3.266	<u>3.939</u>
Qwen3-TTS-VD (Joint)	66.80	66.37	70.66	0.922	3.8 / <u>1.3</u>	3.794	3.952
Qwen3-TTS-VD (TED-TTS)	74.25	74.02	<u>78.03</u>	0.911	<u>4.5</u> / 1.7	<u>3.655</u>	3.853
Qwen3-TTS-VD (EDICT-Stage II)	<u>82.30</u>	<u>83.60</u>	82.00	<u>0.915</u>	5.1 / 1.8	3.631	3.930
CosyVoice 2 (Concat.)	62.43	62.73	31.80	0.917	<u>6.5</u> / <u>3.6</u>	3.916	<u>3.866</u>
CosyVoice 2 (Joint)	49.68	50.19	<u>69.71</u>	0.939	5.4 / 3.1	3.877	3.897
CosyVoice 2 (TED-TTS)	58.10	58.00	63.94	0.920	7.5 / 4.6	4.021	3.828
CosyVoice 2 (EDICT-Stage II)	<u>60.56</u>	<u>60.10</u>	76.20	<u>0.924</u>	8.1 / 5.5	<u>3.978</u>	3.844

Table 4: **Joint timbre and delivery control with Qwen3-TTS-VD.** Requests combine TimbreEdit-Bench voice edits with IntraTTS-Bench texts and delivery instructions. SIM-T compares the complete output with the target voice; SIM-I measures inter-segment speaker similarity. WER/CER reports English WER / Chinese CER in percent. Best and second-best scores are **bold** and underlined, with CER and WER ranked separately.

Configuration	Control (%)		Content and quality			Consistency	SIM-T↑	
	SegAcc↑	Trans↑	WER/CER (%)↓	UTMOS↑	DNSMOS↑		WavLM	ERes2Net
Qwen3-TTS-VD (Joint)	60.19	<u>70.07</u>	3.6 / 1.2	3.339	3.869	0.918	0.683	0.395
Qwen3-TTS-VD (Concat.)	<u>86.43</u>	56.69	<u>4.5</u> / 0.8	<u>3.669</u>	<u>3.895</u>	0.884	0.694	0.405
EDICT Stage I + Concat.	91.17	58.47	4.8 / <u>1.1</u>	3.296	3.739	0.881	<u>0.749</u>	<u>0.598</u>
EDICT	81.62	83.45	5.1 / 1.9	3.795	3.921	<u>0.917</u>	0.788	0.609

improves temporal continuity and speaker consistency, albeit at the cost of lower segment-level accuracy and a higher recognition error rate. The EDICT system achieves the highest target similarity and predicted perceptual quality, while maintaining speaker consistency close to that of *Qwen3-TTS-VD (Joint)*. These results demonstrate that an edited reference can effectively guide local delivery changes within a single utterance, although KV cache reconstruction does not fully preserve the segment-level accuracy achieved by independent synthesis.

5.4 SUBJECTIVE EVALUATION

Figure 4 presents the results of the subjective listening tests. For timbre editing, *EDICT Stage I* achieves the highest mean ratings for edit accuracy and target similarity among instruction-conditioned systems, while maintaining naturalness comparable to *Qwen3-TTS-VD*. Seed-VC attains higher target similarity, but it is conditioned directly on the target audio rather than a textual instruction. For intra-utterance control, *Concat.* performs best in instruction following but receives the lowest transition-quality rating, highlighting the discontinuities introduced by independent segment synthesis. *EDICT Stage II* outperforms TED-TTS in both instruction following and transition quality, while achieving naturalness and transition quality close to those of *Joint*. These subjective ratings draw same conclusions as in the objective evaluations.

5.5 ABLATION STUDIES

Timbre-editor training. Table 5 shows that more SFT data improves target matching for structured instructions, but gives mixed results for descriptive instructions. A larger candidate pool improves DPO’s SIM-T under both instruction forms. wDPO achieves the highest SIM-T, but yields higher Chinese CER than SFT (1M), with mixed changes in predicted speech quality. Appendix H.2 provides further ablations.

KV cache reconstruction. Table 6 compares reconstruction with retaining or truncating the existing cache. Reconstruction improves all three control metrics over both alternatives, indicating that shortening the cache alone does not account for the gains. Removing initial context improves

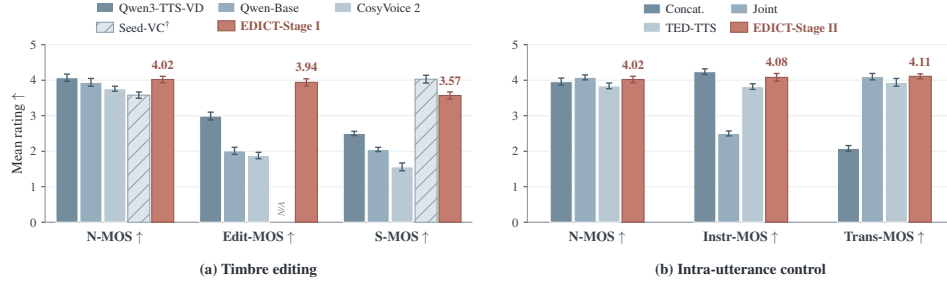


Figure 4: **Subjective evaluation of timbre editing and local control.** Bars show mean five-point ratings from 19 listeners; error bars are computed from 95% bootstrap confidence intervals over audio items. Each system has 10 items in (a) and 12 in (b). EDICT is highlighted in coral; all variants in (b) use Qwen3-TTS-VD. †Seed-VC receives target audio and is not rated for instruction-based edit accuracy.

Table 5: **Timbre-editor training ablations on TimbreEdit-Bench.** All post-training variants start from SFT (1M). M denotes the candidate count per condition; wDPO includes auxiliary SFT as defined in Equation 2. SIM-T measures target-voice similarity; WER/CER reports English WER / Chinese CER in percent. Best and second-best scores within each instruction form are **bold** and underlined, with CER and WER ranked separately.

Configuration	Instruction	WER/CER (%)↓	UTMOS↑	DNSMOS↑	SIM-T↑	
					WavLM	ERes2Net
SFT (0.5M)	Structured	3.9 / <u>1.5</u>	3.689	3.895	0.736	0.498
	Descriptive	4.2 / 1.5	3.708	3.885	0.698	0.481
SFT (1M)	Structured	<u>4.1</u> / 0.9	3.936	3.896	0.757	0.561
	Descriptive	3.9 / 0.9	3.666	<u>3.913</u>	0.693	0.556
DPO ($M = 4$)	Structured	5.1 / <u>1.5</u>	<u>3.885</u>	3.962	0.761	0.552
	Descriptive	4.6 / 1.4	<u>3.768</u>	3.909	0.711	0.543
DPO ($M = 16$)	Structured	4.5 / 1.7	3.786	3.867	<u>0.778</u>	<u>0.626</u>
	Descriptive	4.1 / 1.6	3.724	3.906	<u>0.726</u>	<u>0.634</u>
wDPO ($M = 16$)	Structured	<u>4.1</u> / 1.7	3.830	<u>3.942</u>	0.819	0.653
	Descriptive	<u>4.0</u> / <u>1.1</u>	3.816	3.963	0.793	0.648

Table 6: **KV cache reconstruction ablations on IntraTTS-Bench.** All variants use Qwen3-TTS-VD. The full configuration retains the first $S = 4$ and most recent $K = 5$ acoustic positions; the context ablations remove one of these groups. Persistent and truncated caches reuse old KV states, whereas EDICT recomputes them. Best and second-best scores are **bold** and underlined, with WER and CER ranked separately.

Configuration	Control (%)			Consistency	Content and quality		
	SegAcc↑	EmoAcc↑	Trans↑		WER/CER (%)↓	UTMOS↑	DNSMOS↑
Persistent KV cache	54.25	54.02	68.00	0.918	3.9 / 0.8	3.555	3.953
Truncated KV cache	58.05	56.70	64.94	0.892	4.4 / <u>1.1</u>	3.278	3.905
EDICT-Stage II	<u>82.30</u>	<u>83.60</u>	82.00	<u>0.915</u>	5.1 / 1.8	3.631	<u>3.930</u>
w/o initial context	89.29	89.00	<u>79.99</u>	0.863	4.3 / 1.4	<u>3.585</u>	3.896
w/o recent context	81.15	82.26	55.32	0.875	<u>4.2</u> / <u>1.1</u>	3.548	3.830

local accuracy but weakens speaker consistency; removing recent context sharply degrades transitions. These complementary effects support using initial context for voice stability and recent context for continuation. Reconstruction also produces higher recognition error than the persistent cache, consistent with the cost in content fidelity seen in the main comparison. Appendix H.3 reports context-length sweeps.

6 CONCLUSION

In this paper, we present EDICT, a framework for combining reference-conditioned timbre editing with intra-utterance instruction control in text-to-speech synthesis. An attribute-based timbre editor produces an edited reference that anchors the shared voice, while segment-wise KV cache reconstruction updates local delivery conditions and retains selected acoustic context. We introduce TimbreEdit-Bench and IntraTTS-Bench to evaluate these complementary forms of control. Objective evaluations, subjective listening tests, and ablation studies demonstrate improved timbre-edit accuracy and target-voice matching, alongside a favorable balance between local instruction adherence, voice consistency, and natural transitions. Joint-control experiments further show that edited references can support successive delivery changes within a single utterance. Future work will expand the timbre-editing data to cover more diverse voice characteristics and editing requests, and explore applications in conversational speech generation.

AI USE STATEMENT

We used generative AI tools to assist in constructing synthetic training data and benchmarks, including instruction generation and request parsing, as detailed in Appendix C. Generative AI was also used in the automatic evaluation of generated speech; the corresponding judge prompts, evaluation protocols, and human verification procedures are described in Appendix E. In addition, LLMs were used for language refinement. All generated materials, evaluation procedures, and manuscript content were reviewed and validated by the authors, who take full responsibility for the final content of this work.

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APPENDICES

This appendix provides the task definitions, data construction, implementation details, evaluation protocols, and additional analyses. The contents below link to each section and subsection.

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Table 7: **Taxonomy of speech control tasks.** Typical control inputs, control targets, and temporal scope are shown. Inputs are not exhaustive, and tasks may overlap.

Task	Control input	Control target	Scope
Speech content editing	Recording + revised text	Spoken content	Selected span
Timbre editing	Reference + edit instruction	Selected timbre attributes	Utterance
Voice cloning	Reference speech	Voice identity	Utterance
Voice design	Description / attribute specification	Voice characteristics	Utterance
Global expressive control	Utterance-level instruction	Emotion, rate, prosody, etc	Utterance
Local expressive control	Span-specific instructions	Emotion, rate, prosody, etc	Word / segment

A TASK TAXONOMY AND METHOD COMPARISON

Task distinctions. Table 7 distinguishes six related tasks by their typical inputs, targets, and temporal scope. Speech content editing changes the words in an existing recording; timbre editing changes specified voice attributes while aiming to retain others. Voice cloning and design specify a voice through an acoustic exemplar and a description, respectively. Global and local expressive control apply delivery conditions to an utterance or to individual spans. Instructions are an interface across these tasks, not a separate task.

Scope of the capability comparison. Table 8 records capabilities of the named models in the cited settings. Its timbre-modification column includes both source-voice editing and target-voice conversion; a positive entry does not establish selective preservation of unspecified attributes. Intra-utterance control concerns delivery assigned to words or segments, excluding content edits and paralinguistic insertions. Superscripts identify restricted scope or additional training. EDICT’s voice-cloning and voice-design capabilities are inherited from its backbone and are not evaluated here; this work evaluates reference-conditioned editing and segment-level delivery control. Section 2 discusses the related methods. Additional related methods explore several complementary settings. Reference-conditioned synthesis includes NaturalSpeech 3 (Ju et al., 2024), Seed-TTS (Anastassiou et al., 2024a), and F5-TTS (Chen et al., 2025). FC-TTS and Control-TTS separately control timbre and style through speech references (Lee et al., 2026; Lan et al., 2026), while (Zhao et al., 2025; 2026) supports prosody or pitch control with timbre preservation in voice conversion and synthesis. LoopTTS (Song et al., 2026) corrects local prosodic defects through diagnosis and instruction-conditioned re-synthesis. InstructAudio (Qiang et al., 2026) extends natural-language control to both speech and music generation.

B TIMBRE-EDITING TRAINING DATA

Attribute schema. Table 9 defines the nine attributes used for profile generation and edit encoding in Section 3.2. Categorical edits specify target values, while ordinal edits specify signed level changes. Unchanged attributes receive keep operations.

Speech-bank construction. SpeechCraft (Jin et al., 2024) and CapSpeech (Wang et al., 2026b) provide speech paired with natural-language descriptions. However, these resources primarily describe the characteristics of individual speech samples, whereas our task requires source–target pairs specifying which voice attributes to change and which to preserve. We therefore construct editing supervision from a synthetic speech bank organized by the nine-attribute timbre schema introduced in Section 3.2. For each of 6,000 profiles, Qwen3-14B (Yang et al., 2025a)² produces a natural-language voice description, which we combine with a shared neutral-delivery instruction to reduce variation in non-timbre factors. This instruction specifies a normal speaking rate, medium loudness, and natural pauses, without whispering, shouting, exaggerated acting, or a regional accent. The text

²<https://huggingface.co/Qwen/Qwen3-14B>

Table 8: **Comparison of speech generation and editing methods.** Capabilities refer to the named models in the cited settings. Superscripts specify restricted scope or additional training.

Method	Zero-shot cloning	Voice design	Timbre modification	Intra-utterance control	Control formulation
VoxInstruct (Zhou et al., 2024)	✓	✓	✗	✓ ^a	Instruction-guided synthesis and word-level emphasis
Qwen3-TTS-Base (Hu et al., 2026a)	✓	✗	✗	✗	Reference-based voice cloning
Qwen3-TTS-VoiceDesign (Hu et al., 2026a)	✓	✓	✗	✗	Description-guided voice design
Spark-TTS (Wang et al., 2025b)	✓	✓	✗	✗	Attribute-guided design and reference cloning
VoiceSculptor (Hu et al., 2026b)	✓	✓	✗	✗	Voice design followed by cloning
CosyVoice 2 (Du et al., 2024)	✓	✗	✗	✓ ^b	Instructed cloning and phrase-level tags
IndexTTS2 (Zhou et al., 2026a)	✓	✗	✗	✗	Cloning with global emotion and duration control
FlexiVoice (Chen et al., 2026)	✓	✓ ^c	✗	✗	Instructed generation with optional voice reference
VoxCPM2 (Zhou et al., 2026c)	✓	✓	✗	✗	Voice design and style-controlled cloning
VoxEditor (Sheng et al., 2025)	✗	✗	✓	✗	Relative voice-attribute editing
DreamVC (Hai et al., 2024)	✗	✗ ^d	✓ ^d	✗	Conversion to a text-described target voice
Step-Audio-EditX (Yan et al., 2025)	✓	✗	✓ ^e	✗	Style editing and paralinguistic insertion
FireRedTTS3-Instruct (Shen et al., 2026)	✓	✓	✓ ^f	✗	Voice design; acoustic and content editing
WeSCon (Wang et al., 2025a)	✓	✗	✗	✓	Word-level emotion and rate control
TED-TTS (Liang et al., 2026)	✓	✗	✗	✓	Segment-level emotion and duration control
EDICT	✓ ^g	✓ ^g	✓	✓	Reference timbre editing and segment-level delivery

Legend. ✓: reported within the stated scope; ✗: not explicitly reported in the cited setting. Voice design includes description- or attribute-guided generation without input audio. Timbre modification includes source-voice editing and conversion. Intra-utterance control concerns delivery assigned to words or text spans; vocalization insertion and content edits are excluded.

^a Word-level emphasis after additional fine-tuning.

^b Phrase-level emphasis and laughter tags.

^c Description-conditioned generation without a voice reference.

^d DreamVC only, excluding DreamVG; target-voice conversion does not establish selective attribute preservation.

^e Voice-related style tags, such as older, child, and girl.

^f Timbre-related editing is limited here to pitch alteration.

^g Cloning and voice design are supported by the EDICT system but are not evaluated in this work.

Table 9: **Perceptual attribute schema for timbre editing.** Perceived gender and age impression are categorical; the seven remaining attributes use ordinal levels 1–3 in the listed order.

Attribute	Values	Perceptual description
Perceived gender	Masculine, Feminine	Perceived vocal masculinity or femininity.
Age impression	Child, Young adult, Mature	Perceived age conveyed by the voice.
Pitch register	Low, Mid, High	Typical pitch range of the voice.
Brightness	Dark, Neutral, Bright	Perceived prominence of high-frequency energy.
Vocal weight	Thin, Medium, Thick	Perceived fullness and body of the voice.
Resonance focus	Head, Balanced, Chest	Resonance perceived as head- or chest-focused.
Roughness	Smooth, Moderate, Rough	Perceived irregularity in voicing.
Breathiness	Clear, Slight, Breathy	Audible breath noise during phonation.
Nasality	Oral, Slight, Nasal	Perceived strength of nasal resonance.

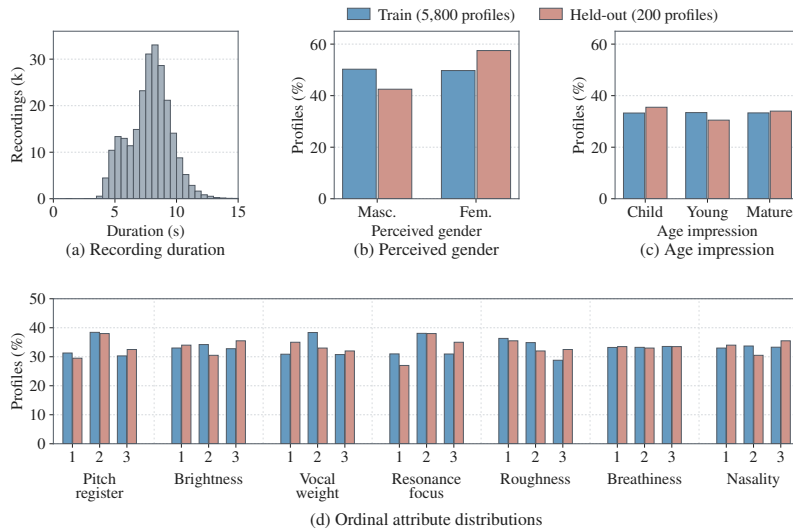


Figure 5: **Synthetic speech-bank statistics.** (a) Retained recording durations in 0.5-second bins. (b,c) Perceived gender and age impression. (d) Distributions of the seven ordinal attributes; levels 1–3 follow the schema in Table 9. Profile percentages are normalized within the training (5,800) and held-out (200) subsets. Young denotes young adult.

pool comprises 1,500 Chinese–English parallel pairs. Each profile is assigned 20 text pairs: five shared across all profiles and 15 selected from the pool. We synthesize both language versions with Qwen3-TTS-VD (Hu et al., 2026a),³ producing 240,000 recordings. Filtering is based on generation status and duration; no ASR-based transcript checks or perceptual attribute screening are applied. Removing 206 recordings longer than 15 seconds leaves 239,794 recordings totaling 518.23 hours, with a mean duration of 7.78 seconds. All durations are derived from codec frame counts at 12 Hz. We reserve 200 profiles for constructing TimbreEdit-Bench and use the remaining 5,800 for training, excluding held-out recordings from both sides of every training pair. Figure 5 summarizes the recording durations and attribute distributions.

Training-pair construction. We form directed editing pairs from recordings associated with different training profiles. The source recording provides the reference voice, while the target recording and its transcript provide synthesis supervision. Edit operations are determined by the paired profiles: signed level differences for ordinal attributes, target categories for changed categorical attributes, and keep operations for unchanged attributes. Reusing recordings across pairs yields one million training examples spanning 826,835 distinct directed profile pairs. Same-text pairs account

³<https://huggingface.co/Qwen/Qwen3-TTS-12Hz-1.7B-VoiceDesign>

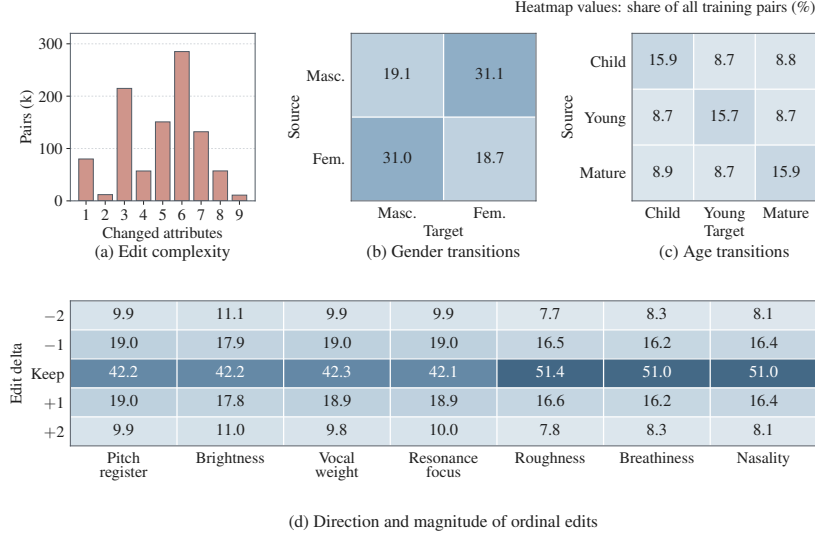


Figure 6: **Edit coverage across one million training pairs.** (a) Number of changed attributes per pair. (b,c) Source-to-target transitions for perceived gender and age impression. (d) Signed target-minus-source differences for ordinal attributes; Keep denotes no change. Heatmaps use a shared color scale and show percentages of all pairs. Each complete matrix in (b,c) and each column in (d) sums to 100% before rounding. Young denotes young adult.

for 51.9% of examples, different-text pairs within a language for 40.9%, and cross-language pairs for 7.2%. The target language is nearly balanced between Chinese (500,236 examples) and English (499,764 examples). Together, the pairs use 229,420 distinct recordings totaling 495.37 hours, counting each recording once across the source and target sets. The combined source and target duration averages 13.44 seconds per pair. Accumulated over all pairs, this amounts to 3,732.51 hours, with reused recordings counted at each occurrence.

Edit coverage. The training pairs cover both single-attribute and compositional edits, including changes of one or two levels in either direction for ordinal attributes. Of the one million pairs, 80,000 change a single attribute and 920,000 change multiple attributes, with an average of 4.92 changed attributes per pair. Every pair specifies at least one change; the training set contains no all-keep examples. Figure 6 shows the distribution of edit complexity, categorical transitions, and ordinal edit operations. These statistics characterize the intended edits derived from profiles; they do not measure how accurately the synthesized audio realizes them.

C BENCHMARK CONSTRUCTION

Figure 3 summarizes the two complementary evaluation tasks.

C.1 TIMBREEDIT-BENCH

Pair and instruction construction. We construct 2,200 candidate editing pairs from the 200 held-out timbre profiles described in Section 4.1. The pairs span nine categories of single-attribute and compositional edits, including age, acoustic, and cross-gender changes. Each example provides source and target recordings, a synthesis text, and two instruction forms. Structured instructions specify profile-derived attribute changes, which an LLM rewrites as natural descriptive requests using the template summarized in Prompt 1. Programmatic checks and a separate LLM verifier assess direction, attribute coverage, intensity, and fluency; rejected candidates are revised with feedback. Each descriptive variant must preserve every requested attribute change, its direction, and its target category or intensity, together with any explicit preservation requirements. The acceptance rule permits no omitted or added attribute operations, reversed directions, or altered target categories or

intensities, regardless of edit complexity. Candidates that violate these requirements are rejected and revised.

Screening and composition. Three annotators screen the candidate pairs. Using the recordings, instruction, and attribute descriptions, they assess whether the requested changes are audible, unmodified attributes remain consistent, and audio quality permits a reliable judgment. The retained benchmark comprises 1,955 pairs: 975 with Chinese speech and 980 with English speech, including 358 single-attribute and 1,597 compositional edits. Source and target recordings average 6.40 and 6.38 seconds, respectively. Prompt 1 is condensed for presentation; the full template is preserved in the construction materials.

C.2 INTRATTS-BENCH

Example construction and validation. We jointly generate a coherent utterance, two or three consecutive text segments, and a delivery instruction for each segment, conditioned on language and topic (dialogue, observation, or vivid description). Prompt 2 summarizes the generation template, which requests natural changes in emotion, rate, pauses, emphasis, intonation, or energy while excluding timbre and speaker-identity edits. Programmatic checks validate metadata, segment counts, nonempty fields, and text lengths, and reject prohibited identity-related terms. They also require the concatenated segments to reproduce the full text after whitespace normalization. These checks are repeated when merging the collection. Shared voice references are assigned separately during synthesis.

Dataset composition. IntraTTS-Bench contains 997 requests (497 Chinese and 500 English), including 598 two-segment and 399 three-segment examples, for a total of 2,393 segments and 1,396 adjacent boundaries. Dialogue, observation, and vivid description account for 333, 334, and 330 requests, respectively. Figure 7 summarizes the request composition and segment lengths. The condensed template in Prompt 2 retains the input fields, generation constraints, and output structure; the complete version is preserved in the construction materials.

Prompt 1. TimbreEdit-Bench: descriptive instruction generation

Input.

Source voice description: {ref_desc}

Target voice description: {tgt_desc}

Structured edit instruction: {canonical}

Task. Rewrite the structured instruction as two natural voice-editing requests in {language}. Use the structured instruction as ground truth: each variant must independently retain every requested attribute change, its direction, and its target category or intensity, together with any explicit preservation requirements. No attribute change may be omitted, regardless of the number of attributes involved. Do not add, reverse, strengthen, or weaken any operation. Use the source and target descriptions only to guide wording, without introducing additional editing requirements. Describe the desired voice naturally rather than listing attribute transitions mechanically. Make clear that the request edits the reference voice. Do not append a boilerplate statement about keeping other traits unchanged unless it is explicitly required by the structured instruction. Each variant must be one sentence, within 80 Chinese characters or 60 English words; shorten unnecessary wording rather than omit required edits. Before returning, check each variant against the structured instruction attribute by attribute and rewrite any variant with missing, added, or altered operations.

Output. Return JSON only:

```
{ "variants": [ "...", "..." ] }
```

Prompt 2. IntraTTS-Bench: segmented text and instruction generation

Input. {language}, {topic}, {topic_definition}, and {num_segments}.

Task. Generate one coherent, speakable sentence on the given topic and divide it into exactly {num_segments} consecutive segments. Assign one concise delivery instruction to each segment, covering emotion, rate, pauses, emphasis, intonation, energy, loudness, or speaking manner. Create clear but natural variation across segments. Do not request changes to timbre, speaker identity, gender, age, or accent. Avoid rare named entities, numbers, URLs, code, brackets, and emoji. Target 18–36 Chinese

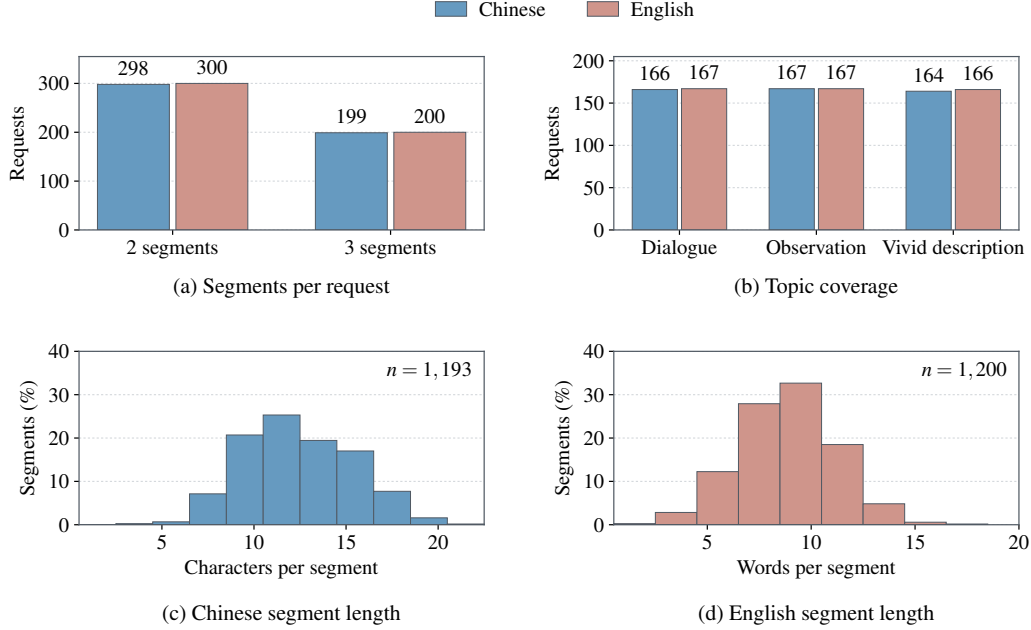


Figure 7: **IntraTTS-Bench composition.** (a,b) Request counts by segment count and topic, split by language. (c,d) Segment lengths in bins of width two, normalized within each language. Chinese lengths count non-whitespace characters, including punctuation; English lengths count whitespace-separated words.

Prompt 2. IntraTTS-Bench: segmented text and instruction generation (continued)

characters or 15–28 English words. Concatenating the segment texts must reproduce the full sentence with its punctuation and order intact.

Output. Return JSON only; the segment list must have the requested length:

```
{
  "language": "...",
  "topic": "...",
  "num_segments": N,
  "text": "...",
  "segments": [
    {
      "text": "...",
      "instruction": "...",
      ...
    }
  ]
}
```

D IMPLEMENTATION DETAILS

D.1 TIMBRE-EDITOR TRAINING

We initialize the timbre editor from Qwen3-TTS-12Hz-1.7B-Base and fine-tune all parameters. SFT runs for 50 epochs on 1M pairs using AdamW with a learning rate of 4×10^{-6} , an effective batch size of 1,024, weight decay of 0.1, and cosine decay with 10% warmup. During initial SFT, the residual-codebook loss has a weight of 0.3, and conditioning dropout is set to 0.1. We then perform one round of wDPO with auxiliary SFT. The frozen SFT model samples $M = 16$ candidates for each of 20K editing conditions drawn from the training pairs, excluding all 200 held-out profiles from both source and target sides. Post-training uses a constant learning rate of 10^{-6} , eight conditions per update, DPO scale $\beta = 0.1$, and auxiliary SFT coefficient $\lambda_{\text{SFT}} = 0.01$. Qwen3-14B parses editing instructions, and a frozen Qwen3-TTS-VD model performs multi-instruction generation.

Preference-pair construction. Preference optimization has been explored for text and speaker fidelity in TTS (Hussain et al., 2025) and for non-verbal vocalization synthesis (Li et al., 2026b). In our method, for each editing condition g , we compute a score $s_{g,i}$ for decoded candidate i as the cosine similarity between its speaker embedding and that of the paired target recording, using a frozen WavLM encoder (Chen et al., 2022). We select the highest-scoring valid candidate $\mathbf{c}_g^+$, with score s_g^+ , and require $s_g^+ \geq 0.70$. An eligible rejected candidate $\mathbf{c}_{g,j}^-$ must be valid and satisfy

$s_{g,j}^- \geq 0.35$ and $\Delta_{g,j} = s_g^+ - s_{g,j}^- \geq 0.15$. We pair each chosen candidate with up to four eligible candidates having the lowest scores. Conditions with no eligible pair are excluded. These preferences are determined by target-speaker similarity and do not directly account for intelligibility or attribute preservation.

Pair weighting and loss computation. Each retained pair is assigned a fixed weight based on its reward gap:

$$w_{g,j} = \text{clip}\left(\frac{\Delta_{g,j}}{0.2}, 0.5, 2.0\right). \quad (5)$$

The preference term in Equation 2 is normalized by the sum of pair weights in each minibatch. We use the frozen SFT sampling model as the reference p_{ref} . For both p_θ and p_{ref} , we evaluate each chosen and rejected codec sequence under teacher forcing with its editing condition g . The auxiliary SFT term applies teacher-forced codec cross-entropy to the chosen sequences under the same conditions, following prior work on supervised regularization for preference optimization (Liu et al., 2024; Li et al., 2026b). The candidate pool, similarity scores, preference pairs, and weights remain fixed throughout post-training.

D.2 SEGMENT-LEVEL DECODING

Generation resumes from the reconstructed cache, which grows as newly generated acoustic frames are fed back into the model. Equation 4 describes attention to the rebuilt prefill and the active segment’s acoustic inputs through the current input position. As illustrated in the upper-right panel of Figure 2, standard causal attention is preserved: subsequent queries access the reconstructed prefill and the growing segment history. Intermediate acoustic history excluded during reconstruction is no longer available in the cache, but its codec frames remain in the output sequence. Non-final $\langle \text{EOS} \rangle$ markers trigger reconstruction without entering the acoustic output, whereas the final $\langle \text{EOS} \rangle$ terminates generation. The codec sequences from successive segments form a single stream that is decoded into the final waveform. Appendix G visualizes attention during cache reconstruction and subsequent decoding.

D.3 BASELINE AND ABLATION CONFIGURATIONS

Baseline implementations.

- **Timbre editing.** Qwen3-TTS-VD and Qwen3-TTS-Base receive the source reference through in-context learning, together with the editing instruction and target text. CosyVoice 2 uses the source recording as its speech prompt and receives the edit through its instruction interface. We evaluate both structured and descriptive instructions. FreeVC and Seed-VC instead receive source and target-reference audio without editing instructions, and are evaluated separately on a same-text subset.
- **Intra-utterance control.** The *Concat.* baseline independently synthesizes each segment with its local instruction and directly concatenates the waveforms, without crossfading or added silence. The *Joint* baseline generates the complete utterance in one pass, with an instruction explicitly mapping delivery conditions to text spans. We adapt TED-TTS to each backbone, using online monotonic alignment to switch segment-specific conditions through attention masking while retaining causal acoustic history, without KV cache reconstruction.
- **Joint control.** *Qwen3-TTS-VD (Joint)* and *Qwen3-TTS-VD (Concat.)* receive the original reference, global edit request, and local delivery instructions. Each *Concat.* request includes the global edit and the corresponding local instruction. Editor + *EDICT Stage 1* + *Concat.* and EDICT use the same precomputed edited reference, followed by independent segment synthesis or continuous generation with cache reconstruction, respectively.

Ablation configurations.

- **Timbre-editor training.** We compare SFT on 0.5M and 1M editing pairs. Both DPO baselines start from the 1M-pair SFT checkpoint, use uniform pair weights with auxiliary SFT, and share the same candidate pool, with $M = 4$ subsampled from $M = 16$. DPO-RP ($M = 16$) uses the

following score for candidate ranking and margin filtering:

$$s_i^{\text{RP}} = s_i^{\text{tgt}} - \lambda \max(0, s_i^{\text{src}} - \tau), \quad \tau = 0.75, \quad \lambda \in \{0.1, 0.2\}, \quad (6)$$

where s_i^{tgt} and s_i^{src} are WavLM speaker-embedding cosine similarities to the target and source, respectively. The eligibility thresholds and standard DPO loss remain unchanged. All DPO and wDPO variants include auxiliary SFT. Learning-rate and training-budget differences prevent isolating the effect of the scoring rule. The additional wDPO round starts from the first-round policy and uses $M = 16$ newly sampled candidates with the target-only reward.

- **KV cache reconstruction.** All variants use Qwen3-TTS-VD. Persistent KV cache retains the previous acoustic KV states across segment switches. Truncated KV cache retains only the initial S and most recent K acoustic positions but reuses their existing KV states. EDICT instead reconstructs the cache under the updated conditions. The full configuration uses $(S, K) = (4, 5)$; removing initial or recent context gives $(0, 5)$ or $(4, 0)$, respectively, while retaining reconstruction.

E AUTOMATIC EVALUATION PROTOCOLS

Objective metrics. We transcribe both languages with Whisper large-v3. Normalization removes punctuation and whitespace for Chinese, and lowercases and removes punctuation for English. Utterance-level CER and WER are averaged separately by language. Speech quality is assessed with UTMOS and DNSMOS (OVRL). Speaker similarity uses WavLM⁴ and ERes2Net⁵. SIM-T/R compare complete generated and reference recordings; joint-control SIM-T uses the paired TimbreEdit-Bench target recording. For SIM-I, we use generation-provided segments when available and forced alignment to segment jointly generated audio. Similarities are averaged over all unordered segment pairs within each utterance, then across utterances.

LLM-based evaluation. We use Gemini 3.1 Pro⁶, with temperature zero, without revealing system identities. For SegAcc and EmoAcc, the judge receives the complete utterance and its ordered segment texts and instructions, and makes separate pass/fail judgments of instruction following and emotion realization. Prompt 3 specifies these criteria and the required output. Emotion-eligible segments are identified from the instructions and held fixed across systems. A separate call assesses speaker continuity and natural connection at each boundary, allowing audible but natural changes in delivery. Prompt 4 gives the boundary evaluation criteria. For attribute-operation evaluation, the judge receives source and generated audio, attribute definitions, and benchmark-annotated operations, but no target audio. Prompt 5 defines separate judgments for ordinal direction, categorical targets, and explicit keep operations. For each operation type, pass/fail outcomes are averaged over applicable attributes within a request, then across eligible requests. The condensed templates below summarize the evaluation criteria.

Prompt 3. Segment-level instruction and emotion evaluation

Input. One complete generated utterance; {ordered segment texts and instructions}; {emotion-eligible segment indices}.

Task. Evaluate each segment against its own instruction. Judge separately whether the delivery matches all explicitly requested attributes and whether it conveys the requested emotion. Assess emotion independently of other requirements, such as speaking rate; use `not_applicable` for emotion-ineligible segments. Do not impose unspecified requirements or use speaker identity, recording quality, or pronunciation as separate scoring criteria.

Output. Return JSON with each segment’s index, separate instruction and emotion verdicts (pass/fail, with `not_applicable` allowed for emotion), and brief reasons.

⁴<https://huggingface.co/microsoft/wavlm-base-plus-sv>

⁵https://modelscope.cn/models/iic/speech_eres2net_sv_zh-cn_16k-common

⁶<https://ai.google.dev/gemini-api/docs/models>

Table 10: **Human–Gemini agreement on intra-utterance control.** Valid counts refer to segment-level judgments for SegAcc and EmoAcc, and boundary-level judgments for Trans. Agreement (%) uses jointly valid items for each pair.

Criterion	Valid annotations			Agreement (%)		
	H1	H2	Gemini	H1–H2	H1–Gemini	H2–Gemini
SegAcc	197	198	200	90.4	83.8	85.9
EmoAcc	197	198	200	85.8	83.2	85.4
Trans	99	100	100	91.9	93.9	96.0

Table 11: **Human–Gemini agreement on timbre editing.** Each annotator provides 100 valid judgments per criterion. Agreement (%) is computed from paired pass/fail labels.

Criterion	Valid annotations			Agreement (%)		
	H1	H2	Gemini	H1–H2	H1–Gemini	H2–Gemini
Up	100	100	100	81.0	79.0	76.0
Down	100	100	100	73.0	67.0	66.0
Category	100	100	100	82.0	84.0	78.0
Preservation	100	100	100	74.0	72.0	70.0

Prompt 4. Transition evaluation

Input. One complete generated utterance; {ordered segment texts and instructions}.

Task. For each adjacent boundary, judge whether both sides sound like the same speaker and connect naturally, without an audible splice, unnatural gap, or abrupt acoustic reset. Natural pauses and smooth changes in delivery are acceptable; the boundary need not be imperceptible. Changes in pitch, energy, or rate do not by themselves indicate a speaker change. Assess instruction following separately.

Output. Return JSON with the boundary indices, a Boolean judgment for speaker continuity and natural connection, and a combined pass when both are satisfied (fail otherwise), with a brief reason.

Prompt 5. Timbre attribute-operation evaluation

Input. Clip A: source audio; Clip B: generated audio; {attribute definitions with ordered levels}; {required operations}.

Task. Evaluate each attribute operation independently. For *up/down*, compare Clip B with Clip A and judge the requested direction along the supplied ordered levels, not exact step size. For *category*, judge whether Clip B matches the requested perceived-gender or age category. For *keep*, judge whether the attribute remains essentially unchanged, allowing minor natural variation. Do not use linguistic content, pronunciation, recording volume, or overall naturalness as evidence of timbre editing success.

Output. Return JSON with each attribute’s name, required operation, a pass/fail verdict indicating whether it is satisfied, and a brief reason.

Human verification of audio-LLM judgments. Two human annotators (H1 and H2) independently assess the same items using the binary criteria applied to Gemini. For instruction following and emotion realization, Gemini provides 200 valid segment-level judgments per criterion, while H1 and H2 provide 197 and 198, respectively. For transitions, Gemini and H2 each provide 100 valid boundary judgments, and H1 provides 99. Pairwise agreement is computed only over items with valid labels from both members of the corresponding pair. As shown in Table 10, human–Gemini agreement ranges from 83.2% to 96.0%, with the highest agreement on transition quality. For timbre editing, all three annotators provide 100 valid judgments per criterion, covering edit direction, categorical targets, and attribute preservation. Table 11 reports pairwise agreement for these criteria. Agreement is lower for downward edits and preservation in both human–human and human–Gemini comparisons. This pattern suggests that these fine-grained timbre judgments are less consistent than the evaluated intra-utterance criteria.

EDICT PAPER LISTENING STUDY

Study Instructions

You will listen to generated speech samples and rate them using 1–5 integer scores. The survey is split into two separate sections: timbre editing first, then intra-utterance expression control. System names are hidden during rating. Please use headphones, sit in a quiet environment, and keep a comfortable volume.

- ✓ Use the full 1–5 scales shown on screen; do not convert ratings into pass/fail.
- ✓ If an audio file cannot be played, leave the score blank and notify the study coordinator.
- ✓ You may go back, replay audio, and revise ratings before exporting.
- ✓ Your anonymous participant ID is the only identifier collected by this local survey.

▼ Volume test

Before starting, play any sample later and adjust your volume to a comfortable level. Keep the same volume throughout the survey.

(a) General instructions

SECTION 1
Timbre Editing

You will evaluate generated speech for timbre-editing requests. Each candidate is anonymous.

Step 1
Listen to Generated and rate naturalness (MOS).

Step 2
For H-Edit, listen to Source and Generated. Judge whether Generated changes from Source in the way described by the edit goal.

Step 3
Compare Target and Generated for timbre similarity (S-MOS).

In H-Edit, Source is the starting voice and Generated is the edited result. Target audio is hidden until H-Edit is completed for instruction-based systems.

50 timbre candidates are assigned in this section.

SECTION 2
Intra-utterance Expression Control

You will evaluate complete generated utterances with multiple text segments and segment-level expression instructions.

MOS
Rate the naturalness of the full audio.

H-Instr
Judge whether each segment follows its own instruction in the right order.

H-Trans
Judge whether it sounds like the same speaker naturally changing expression.

Use the full audio. Jump buttons appear only when trusted boundary timestamps are available.

48 intra-utterance candidates are assigned in this section.

(b) Timbre editing

(c) Intra-utterance control

Figure 8: **Instructions shown to listening-test participants.** Panel (a) gives general guidance on playback and rating; panels (b) and (c) describe the timbre-editing and intra-utterance control tasks, respectively. System identities are hidden.

F SUBJECTIVE LISTENING TEST

Survey procedure. The survey covers timbre editing and intra-utterance instruction control, with system identities hidden from participants. The study involves 19 participants. For timbre editing, we evaluate 10 generated audio samples per system, totaling 50 samples across five systems. For intra-utterance control, we evaluate 12 samples per system, totaling 48 samples across four systems. All criteria use integer ratings from 1 to 5, with higher scores indicating better performance. Participants review the general and task-specific guidelines shown in Figure 8. They use headphones in a quiet environment at a comfortable, consistent volume. They may replay recordings and revise their responses. No rating is selected by default; playback failures are left unrated. We report mean item-level listener averages with 95% percentile intervals from 10,000 item-bootstrap resamples. Symmetric error bars use the larger deviation from the mean to either interval bound.

Timbre editing. Panel (a) of Figure 9 shows the three-step rating interface. Participants first assess generated-speech naturalness, then compare the source and generated voices against the editing instruction. The target reference is revealed after edit accuracy has been rated, allowing target similarity to be assessed separately. The three criteria are:

- **Naturalness (N-MOS).** Rate the naturalness of the generated speech, considering pronunciation, rhythm, and synthesis artifacts, independently of personal voice preference or instruction following. The scale ranges from 1: *very unnatural*, 2: *somewhat unnatural*, 3: *fair*, 4: *mostly natural*, to 5: *very natural*.
- **Edit accuracy (Edit-MOS).** Judge how well the generated voice realizes the requested changes relative to the source, including explicit preservation requirements. Audio quality and target similarity are assessed separately. The scale ranges from 1: *almost not completed*, 2: *slightly completed*, 3: *partially completed*, 4: *mostly completed*, to 5: *fully completed*.

Step 1: Naturalness

Generated

0:00 / 0:00

MOS: Naturalness Integer 1–5, higher is better

How natural does this generated speech sound?

Rate the naturalness of the speech itself.

► Full scale definitions

1
Very unnatural

2
Somewhat unnatural

3
Fair

4
Mostly natural

5
Very natural

Step 2: Edit instruction completion

Listen to Source first, then Generated. The question is whether Generated successfully changes the Source voice according to the edit goal below.

Source

0:00 / 0:00

Edit goal Source voice → Generated voice

Make it a female voice with medium brightness, medium weight, slight nasality, and a rougher texture.

Generated

0:00 / 0:00

H-Edit: Edit instruction completion Integer 1–5, higher is better

Relative to the source reference speech, to what extent does the generated speech complete the timbre changes requested by the edit instruction?

Use Source as the starting point and judge whether the requested timbre edit was completed.

► Full scale definitions

1
Almost not completed

2
Slightly completed

3
Partially completed

4
Mostly completed

5
Fully completed

3 – There is a clear correct change, but important requirements are still missing or the degree of change is clearly off.

Step 3: Target timbre similarity

Target

0:00 / 0:00

Generated

0:00 / 0:00

S-MOS: Target timbre similarity Integer 1–5, higher is better

Ignoring spoken content and expression, how similar is the generated speech timbre to the target speech timbre?

Compare Generated with Target for speaker/timbre characteristics only.

► Full scale definitions

1
Very different

2
Somewhat dissimilar

3
Moderately similar

4
Quite similar

5
Very similar

Step 1: Naturalness of the full generated utterance

Full Generated Audio

0:00 / 0:00

MOS: Naturalness Integer 1–5, higher is better

How natural does this generated speech sound?

Rate the naturalness of the speech itself.

► Full scale definitions

1

2

3

4

5

Step 2: Segment instructions and transition quality

Global voice condition
用一位年轻女性的声音说话。

Segment	Text	Instruction	Navigation
1	今天的天空灰蒙蒙的，心里也跟着沉重起来。	用伤感的语气说话。	Full audio
2	但是！刚刚收到录取通知书了！太开心了！	突然变得非常开心和兴奋！	Jump here

H-Instr: Instruction following in the full utterance Integer 1–5, higher is better

To what extent does the full utterance satisfy the corresponding expression instruction for each segment in order?

Judge whether each segment follows its own expression instruction at the correct text position.

► Full scale definitions

1

2

3

4

5

H-Trans: Voice continuity and expression transition Integer 1–5, higher is better

While maintaining the same speaker voice characteristics, does the speech show the requested expression changes between segments and transition naturally?

Judge whether it sounds like the same speaker naturally changing expression as required.

► Full scale definitions

1

2

3

4

5

(a) Timbre editing

(b) Intra-utterance control

Figure 9: **Listening-test interfaces.** Panel (a) collects naturalness, edit accuracy, and target timbre similarity ratings in sequence; the target reference is revealed only after edit accuracy is rated. Panel (b) collects naturalness, instruction following, and transition quality ratings for the complete utterance. All criteria use five-point scales.

- **Target timbre similarity (S-MOS).** Compare the generated and target voices, focusing on timbre rather than linguistic content, expression, or recording quality. The scale ranges from 1: *very different*, 2: *somewhat dissimilar*, 3: *moderately similar*, 4: *quite similar*, to 5: *very similar*.

Target-reference voice conversion is rated for naturalness and target similarity only.

Intra-utterance instruction control. Participants listen to the complete generated utterance while reviewing its ordered segment texts and delivery instructions. Panel (b) of Figure 9 shows this interface. Naturalness uses the same N-MOS scale. Two additional criteria receive one rating each for the full utterance:

- **Instruction following (Instr-MOS).** Judge whether the requested delivery attributes, such as emotion, speaking rate, and prosody, are realized at the corresponding text spans. An expression appearing in the wrong segment does not fulfill the instruction. The scale ranges from 1: *most instructions unmet or misplaced*, 2: *only a small part followed*, 3: *partially followed with important mismatches*, 4: *mostly followed with minor deviations*, to 5: *all segments follow their instructions*.

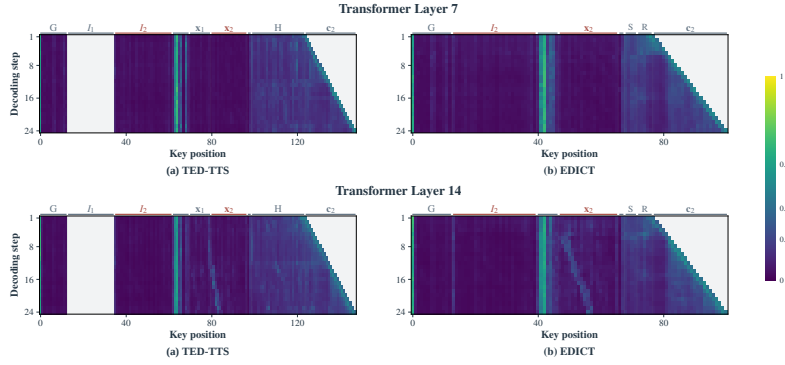


Figure 10: **Attention across segment transitions.** Rows show zero-indexed Transformer layers 7 and 14; columns compare TED-TTS and EDICT on the same request, using separate rollouts. We average attention over all 16 heads in each layer. G denotes the global instruction, I_k/x_k the segment instruction/text, H the preceding audio, S/R the initial/recent context, and c_2 the new segment’s audio positions. Gray cells are masked or unavailable. All panels share a power color scale with $\gamma = 0.35$.

- **Transition quality (Trans-MOS).** Judge whether the utterance sounds like the same person naturally changing expression as instructed. High ratings require both voice continuity and appropriate expression changes; unchanged delivery alone is insufficient. Natural pauses and instruction-appropriate changes in pitch, energy, or rate are acceptable, whereas audible splices, unnatural gaps, truncation, and apparent speaker changes lower the rating. The scale ranges from 1: *severe continuity or transition failures*, 2: *clear problems*, 3: *noticeable flaws*, 4: *natural transitions with minor flaws*, to 5: *smooth, appropriate expression changes with a consistent voice*.

Throughout the paper, we use N-MOS, Edit-MOS, Instr-MOS, and Trans-MOS. The original survey interface labels these criteria MOS, H-Edit, H-Instr, and H-Trans, respectively.

G ATTENTION ANALYSIS OF KV CACHE RECONSTRUCTION

Attention across segment transitions. Figure 10 compares TED-TTS and EDICT decoding on the same two-segment request. We visualize Transformer layers 7 and 14 (zero-indexed), averaging attention weights over all 16 heads within each layer. In TED-TTS decoding, the previous segment’s instruction is masked, while preceding acoustic states remain in the cache. EDICT instead constructs a new context from the updated instruction and text, together with selected initial and recent acoustic inputs. The maps show subsequent decoding attention over this reconstructed context and newly generated audio positions. This distinction concerns both context selection and representation computation: masking old instructions does not itself recompute the acoustic states previously conditioned on them.

Attention during cache reconstruction. Figure 11 shows the complete prefill alongside subsequent decoding. The dashed box marks retained acoustic input positions, whose representations are recomputed under the new segment conditions as part of the full prefill. Subsequent decoding attends to the rebuilt context and newly appended audio positions. The accompanying replay panels contrast old KV under old conditions with reconstruction under updated conditions. This comparison is diagnostic: the old-KV branch receives no new segment instruction or text, and only codec0 inputs, rather than all residual codebooks, are fixed across branches. These visualizations illustrate the computation underlying our method; the quantitative cache ablations in Section 5.5 assess its effects on generation quality and control.

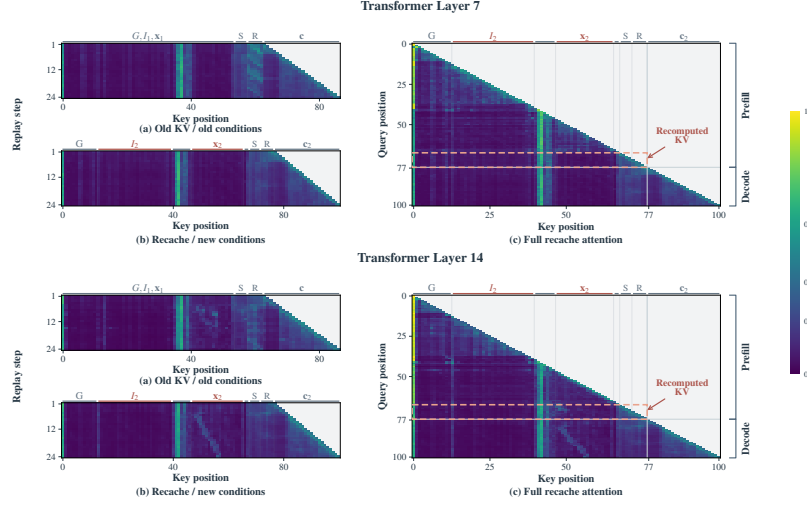


Figure 11: **Attention during KV cache reconstruction.** Each layer group shows replay with (a) old KV states and old conditions, (b) reconstructed KV states and new conditions, and (c) the full prefill followed by decoding. The dashed coral box marks retained acoustic inputs whose KV states are recomputed; solid lines separate prefill from decoding. Layers 7 and 14 are zero-indexed, with attention averaged over all 16 heads. The replay comparison changes both cache and conditions, so it does not isolate reconstruction alone.

H SUPPLEMENTARY EXPERIMENTAL RESULTS

This section provides additional experimental results, with audio demonstrations available at <https://danny-nus.github.io/EDICT-demopage/>.

H.1 TIMBRE EDITING WITH REAL SPEECH REFERENCE

Since both our timbre editor’s training data and the reference voices in TimbreEdit-Bench are synthetic, we construct an additional evaluation set from human-recorded ESD dataset Zhou et al. (2022) to assess whether the learned editing capability extends to real speech references. This set focuses on real human voice compositional edits involving changes to multiple timbre attributes.

Evaluation set construction. We use neutral recordings from 20 Mandarin and English speakers in ESD and manually annotate each speaker using our nine-attribute timbre representation. To construct compositional editing requests, we select speaker pairs with large differences in their annotated timbre attributes. Each case pairs recordings of the same sentence spoken by two different speakers in the same language. Editing requests are derived from the source–target annotation differences, specifying which attributes to modify and which to preserve. The resulting set contains 200 pairs, evenly divided between Mandarin and English, with its statistics information summarized in Figure 12.

Results and discussion. Table 12 shows that *EDICT-Stage 1* consistently outperforms the instruction-conditioned baselines in target-speaker similarity across languages and instruction forms, with agreement between the two speaker encoders. Among these systems, only *EDICT-Stage 1* achieves positive target–source similarity margins in every condition, indicating that its outputs are, on average, closer to the target than to the source in speaker-embedding space. This advantage persists under descriptive instructions, despite lower target similarity than under structured instructions. *EDICT-Stage 1* also achieves the highest DNSMOS in three of the four language–instruction conditions. These results support the transfer of timbre-editing capability from synthetic training data to real speech references on this evaluation set.

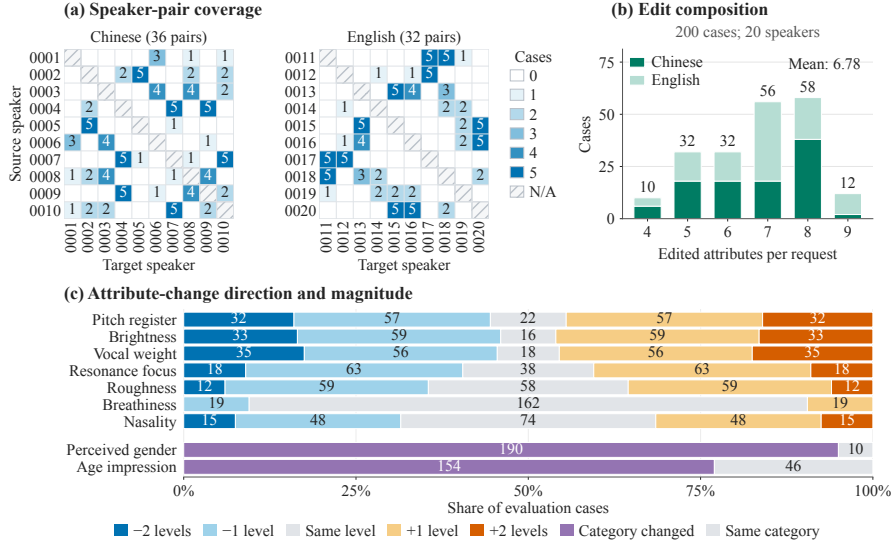


Figure 12: **Statistics of the ESD evaluation set.** (a) Speaker-pair coverage. (b) Number of edited attributes per request. (c) Attribute-change distributions. In-bar numbers indicate case counts.

H.2 TIMBRE EDITING WITH TIMBREEDIT-BENCH

Attribute-level editing and preservation. To complement the target-voice similarity results in Section 5.1, we evaluate whether individual attributes change as requested or remain unchanged. On a fixed subset of 500 TimbreEdit-Bench requests, Gemini assigns pass/fail judgments to applicable attribute operations. Up/Down accuracy assesses ordinal edit direction rather than magnitude, Category accuracy assesses the requested gender and age categories, and Preservation assesses attributes marked as unchanged. Scores are averaged over applicable operations within each request and then across eligible requests. As shown in Table 2, *EDICT-Stage I* achieves the highest Up, Down, and Category accuracy under both instruction forms. Qwen3-TTS-VD obtains the highest preservation scores but lower edit accuracy, highlighting that preservation alone does not establish successful editing. *EDICT-Stage I*’s remaining preservation gap indicates that accurately applying requested changes does not yet guarantee that all other attributes remain unaffected.

Extended timbre-editor ablations. Table 13 examines SFT data scale, post-training objectives, preference scoring, and an additional optimization round. More SFT data improves target similarity for structured instructions but gives mixed results for descriptive instructions. A larger candidate pool improves DPO’s target matching, while wDPO achieves the highest SIM-T among single-round configurations across both instruction forms and speaker encoders. Compared with DPO ($M = 16$), DPO-RP improves WavLM similarity, but its ERes2Net results are mixed. Both RP variants remain below wDPO; differences in pair weighting, learning rate, and training budget prevent attributing this gap solely to the source-similarity penalty. A second wDPO round reduces English WER and improves descriptive WavLM similarity, but reduces ERes2Net similarity and predicted speech quality under both instruction forms. We therefore retain single-round wDPO as the default to balance target matching and speech quality. We examine the auxiliary SFT coefficient below.

Effect of auxiliary SFT regularization. We vary the auxiliary SFT coefficient $\lambda_{\text{SFT}} \in \{0, 0.01, 0.05, 0.20\}$ in wDPO; zero removes the auxiliary SFT term. As shown in Figure 13, $\lambda_{\text{SFT}} = 0.01$ achieves the highest WavLM target-voice similarity under both instruction forms and the highest ERes2Net similarity for descriptive instructions. Larger coefficients can reduce English WER and improve UTMOS, but do not consistently improve target matching. We use $\lambda_{\text{SFT}} = 0.01$ as the default to prioritize target-voice similarity for downstream synthesis, accepting higher recognition error than the lowest-error configurations.

Table 12: **Objective results on the ESD evaluation set.** WER/CER reports English WER or Mandarin CER. SIM-T/R denotes target/source similarity; $\Delta = \text{SIM-T} - \text{SIM-R}$. Best and second-best scores within each language and instruction form are **bold** and underlined; SIM-R is unranked. Target-reference VC is shown separately, with its best similarity and Δ scores in **bold**.

System	Instruction	Content and quality			WavLM			ERes2Net		
		WER/CER (%)↓	UTMOS↑	DNSMOS↑	SIM-T↑	SIM-R	Δ ↑	SIM-T↑	SIM-R	Δ ↑
(a) English										
Instruction-conditioned systems										
Qwen3-TTS-VD	Structured	<u>5.3</u>	<u>4.219</u>	3.487	<u>0.510</u>	0.714	<u>−0.204</u>	<u>0.383</u>	0.531	<u>−0.148</u>
	Descriptive	6.1	<u>4.162</u>	3.400	<u>0.483</u>	0.709	<u>−0.226</u>	<u>0.385</u>	0.533	<u>−0.148</u>
Qwen3-TTS-Base	Structured	4.7	4.358	3.387	0.396	0.934	−0.538	0.301	0.770	−0.469
	Descriptive	5.3	4.345	3.421	0.389	0.948	−0.559	0.297	0.787	−0.490
CosyVoice 2	Structured	6.5	3.992	<u>3.607</u>	0.446	0.792	−0.346	0.271	0.745	−0.474
	Descriptive	8.6	4.085	3.595	0.467	0.809	−0.342	0.279	0.784	−0.505
EDICT-Stage I	Structured	<u>5.3</u>	4.142	3.639	0.767	0.491	0.276	0.594	0.330	0.264
	Descriptive	<u>5.6</u>	4.138	<u>3.539</u>	0.728	0.486	0.242	0.547	0.326	0.221
Target-reference voice conversion										
FreeVC	N/A	12.4	4.243	3.602	0.820	0.525	0.295	0.643	0.316	0.327
Seed-VC	N/A	8.4	4.138	3.688	0.902	0.488	0.414	0.798	0.330	0.468
(b) Mandarin										
Instruction-conditioned systems										
Qwen3-TTS-VD	Structured	2.1	<u>3.867</u>	3.590	<u>0.530</u>	0.763	<u>−0.233</u>	<u>0.481</u>	0.616	<u>−0.135</u>
	Descriptive	<u>2.1</u>	<u>3.874</u>	3.561	<u>0.515</u>	0.758	<u>−0.243</u>	<u>0.485</u>	0.614	<u>−0.129</u>
Qwen3-TTS-Base	Structured	<u>2.4</u>	3.938	3.466	0.426	0.970	−0.544	0.269	0.770	−0.501
	Descriptive	1.9	3.956	3.428	0.411	0.968	−0.557	0.306	0.774	−0.468
CosyVoice 2	Structured	6.3	3.657	<u>3.712</u>	0.444	0.901	−0.457	0.244	0.742	−0.498
	Descriptive	6.7	3.701	<u>3.723</u>	0.421	0.939	−0.518	0.245	0.774	−0.529
EDICT-Stage I	Structured	3.5	3.857	3.726	0.798	0.552	0.246	0.583	0.269	0.314
	Descriptive	3.8	3.849	3.727	0.762	0.555	0.207	0.536	0.271	0.265
Target-reference voice conversion										
FreeVC	N/A	2.4	3.795	3.761	0.862	0.482	0.380	0.631	0.293	0.338
Seed-VC	N/A	1.3	3.361	3.855	0.947	0.475	0.472	0.809	0.277	0.532

H.3 INTRA-UTTERANCE INSTRUCTION CONTROL

Effect of acoustic context length. Tables 14 and 15 examine the effects of initial and recent acoustic context, respectively, on a 500-request subset of IntraTTS-Bench. We measure speaker consistency using SIM-I and emotional alignment using EmoSim, the cosine similarity between emotion2vec embeddings Ma et al. (2024) of each generated segment and a reference segment independently synthesized by Qwen3-TTS-VD from the same text and local instruction. Segment scores are averaged within each utterance and then across utterances. With $K = 5$, increasing S generally improves speaker consistency but reduces EmoSim. With $S = 4$, SIM-I largely plateaus beyond $K = 4$, while larger context windows tend to yield lower EmoSim. These trends suggest a trade-off between voice consistency and alignment with the independently synthesized emotional references, with $(S, K) = (4, 5)$ providing a compromise.

Evaluation on MED-TTS. We extend the evaluation in Section 5.2 to 500 examples from MED-TTS (Liang et al., 2026),⁷ using their segment texts and emotion instructions. Following TED-TTS’s text-prompt protocol, **Emo2v** measures segment-level emotion classification accuracy with the pre-trained emotion2vec+ large model (Ma et al., 2024). After ASR alignment, predicted emotions are compared with target labels over the protocol’s five emotion categories. Emo2v reports percentage

⁷<https://huggingface.co/datasets/Chanson-0803/MED-TTS>

Table 13: **Extended timbre-editor training ablations on TimbreEdit-Bench.** SIM-T measures target-voice similarity; WER/CER reports English WER / Chinese CER in percent. Best and second-best scores within each instruction form are **bold** and underlined, with CER and WER ranked separately.

Configuration	Instruction	Content and quality			WayLM	ERes2Net
		WER/CER (%)↓	UTMOS↑	DNSMOS↑	SIM-T↑	SIM-T↑
<i>(a) SFT data scale</i>						
SFT (0.5M)	Structured	3.9 / 1.5	3.689	3.895	0.736	0.498
	Descriptive	4.2 / 1.5	3.708	3.885	0.698	0.481
SFT (1M)	Structured	4.1 / 0.9	3.936	3.896	0.757	0.561
	Descriptive	<u>3.9</u> / 0.9	3.666	<u>3.913</u>	0.693	0.556
<i>(b) Post-training from SFT 1M</i>						
DPO ($M = 4$)	Structured	5.1 / 1.5	<u>3.885</u>	3.962	0.761	0.552
	Descriptive	4.6 / 1.4	<u>3.768</u>	3.909	0.711	0.543
DPO ($M = 16$)	Structured	4.5 / 1.7	3.786	3.867	0.778	0.626
	Descriptive	4.1 / 1.6	3.724	3.906	0.726	<u>0.634</u>
wDPO ($M = 16$)	Structured	4.1 / 1.7	3.830	<u>3.942</u>	0.819	0.653
	Descriptive	4.0 / <u>1.1</u>	3.816	3.963	<u>0.793</u>	0.648
<i>(c) Preference-score variants</i>						
DPO-RP ($\lambda = 0.1$)	Structured	4.6 / 1.9	3.764	3.880	0.798	0.631
	Descriptive	4.1 / 1.8	3.625	3.891	0.746	0.614
DPO-RP ($\lambda = 0.2$)	Structured	4.4 / <u>1.3</u>	3.688	3.877	0.783	0.616
	Descriptive	<u>3.9</u> / 1.2	3.722	3.908	0.757	0.592
<i>(d) Additional optimization round</i>						
wDPO (round 2)	Structured	3.8 / 1.5	3.754	3.897	<u>0.806</u>	<u>0.645</u>
	Descriptive	3.7 / 1.3	3.653	3.880	0.812	0.631

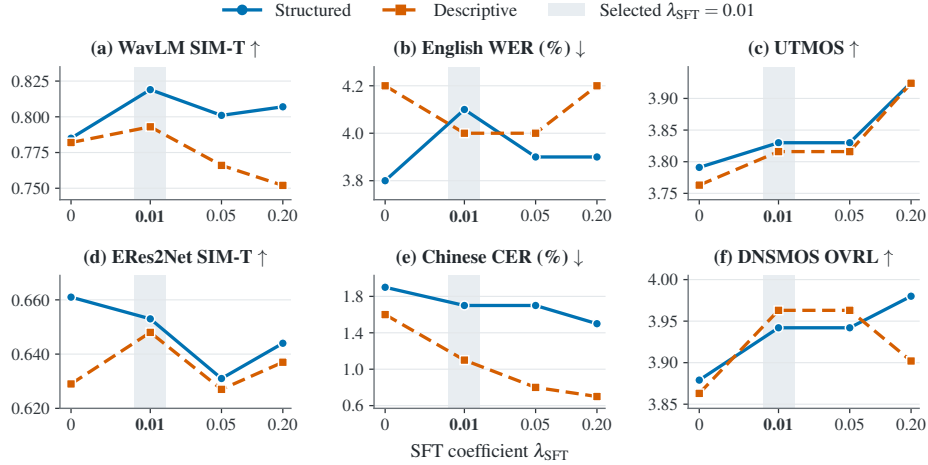


Figure 13: **Sensitivity to the auxiliary SFT coefficient in wDPO on TimbreEdit-Bench.** Solid and dashed curves denote structured and descriptive instructions, respectively. Shading marks the default $\lambda_{\text{SFT}} = 0.01$. The horizontal axis is linear up to 0.01 and logarithmic thereafter.

accuracy, unlike the embedding similarity measured by EmoSim. Table 16 compares *Concat.*, *Joint*, TED-TTS, and EDICT under this protocol. To additionally assess local acoustic continuity, we report **Transition Cepstral Discontinuity** (TCD), a reference-free boundary metric. For each internal boundary b , we average MFCC vectors within the 250 ms windows immediately before and after it and compute

$$\text{TCD}_b = \frac{10\sqrt{2}}{\ln 10} \left\| \bar{\mathbf{m}}_b^{\text{left}} - \bar{\mathbf{m}}_b^{\text{right}} \right\|_2, \quad (7)$$

Table 14: **Effect of initial acoustic context on a 500-request IntraTTS-Bench subset.** With Qwen3-TTS-VD and $K = 5$, we vary the number S of initial acoustic positions retained during reconstruction. SIM-I measures speaker consistency; EmoSim measures emotion-embedding similarity to independently synthesized reference segments. Bold marks the default configuration, rather than the highest score.

Initial context S	0	1	2	3	4	5
SIM-I $\uparrow$	0.872	0.885	0.902	0.899	0.912	0.915
EmoSim $\uparrow$	0.856	0.850	0.842	0.837	0.836	0.821

Table 15: **Effect of recent acoustic context on a 500-request IntraTTS-Bench subset.** With Qwen3-TTS-VD and $S = 4$, we vary the number K of recent acoustic positions retained during reconstruction. SIM-I measures speaker consistency; EmoSim measures emotion-embedding similarity to independently synthesized reference segments. Bold marks the default configuration, rather than the highest score.

Recent context K	1	2	3	4	5	6	7	8	9	10
SIM-I $\uparrow$	0.876	0.895	0.903	0.911	0.912	0.912	0.915	0.917	0.916	0.912
EmoSim $\uparrow$	0.837	0.842	0.838	0.832	0.836	0.831	0.825	0.819	0.821	0.817

Table 16: **Intra-utterance emotion control on 500 MED-TTS examples.** WER/CER denotes English WER or Chinese CER in percent. Emo2v is a segment-level emotion classification accuracy in percent; SIM-I measures inter-segment speaker similarity. TCD is reported as 10^{-3} TCD, and OVRL denotes DNSMOS OVRL. Best and second-best scores within each language are **bold** and underlined.

Language	System	WER/CER $\downarrow$	Emo2v $\uparrow$	SIM-I $\uparrow$	TCD $\downarrow$	OVRL $\uparrow$	UTMOS $\uparrow$
English	Concat.	1.18	88.33	0.667	0.467	3.799	3.550
	Joint	0.30	59.47	<u>0.929</u>	0.487	<u>3.754</u>	4.040
	TED-TTS	<u>0.69</u>	80.91	0.919	<u>0.463</u>	3.676	<u>3.875</u>
	EDICT-Stage II	1.09	<u>85.81</u>	0.932	0.357	3.739	3.863
Chinese	Concat.	<u>1.74</u>	85.56	0.589	0.488	3.665	3.528
	Joint	1.57	61.48	0.856	<u>0.425</u>	<u>3.692</u>	3.763
	TED-TTS	2.68	76.32	<u>0.843</u>	0.449	3.698	3.698
	EDICT-Stage II	2.47	<u>79.96</u>	0.838	0.392	3.679	<u>3.721</u>

where the two vectors represent the mean MFCCs on either side. We average scores over evaluated boundaries and report 10^{-3} TCD. Lower values indicate smaller local spectral changes; this is a continuity proxy rather than a direct measure of transition naturalness, since phonetic content and intended expression changes also affect the score. The aggregate results show the same trade-off as IntraTTS-Bench: *Concat.* provides the strongest emotion control but low speaker similarity, while *Joint* favors speaker consistency. *EDICT-Stage II* improves emotion accuracy over TED-TTS in both languages with comparable speaker similarity. Chinese CER decreases, but English WER increases. EDICT also obtains the lowest TCD in both languages. Table 17 shows that its English emotion gains occur in Emotional Dialogue and Novelistic Description; TED-TTS remains slightly stronger on Observational Narrative. *EDICT-Stage II* has the lowest TCD in all three categories of both languages, indicating smaller boundary spectral changes.

Table 17: **MED-TTS results by language and text category.** WER/CER denotes English WER or Chinese CER in percent. Emo2v is a segment-level emotion classification accuracy in percent; SIM-I measures inter-segment speaker similarity. TCD is reported as 10^{-3} TCD, and OVRL denotes DNSMOS OVRL. Best and second-best scores within each language and category are **bold** and underlined.

	Text category	System	WER/CER↓	Emo2v↑	SIM-I↑	TCD↓	OVRL↑	UTMOS↑
English	Emotional Dialogue	Concat.	2.93	90.18	0.659	<u>0.452</u>	3.807	3.462
		Joint	<u>0.46</u>	65.22	0.912	<u>0.463</u>	<u>3.708</u>	3.752
		TED-TTS	0.34	84.58	<u>0.913</u>	0.487	3.622	3.719
		EDICT-Stage II	1.23	<u>88.58</u>	0.927	0.342	3.699	<u>3.750</u>
	Observational Narrative	Concat.	<u>0.43</u>	84.62	0.652	<u>0.469</u>	3.820	3.525
		Joint	0.12	58.68	0.939	0.508	<u>3.818</u>	4.197
		TED-TTS	1.00	<u>82.69</u>	0.921	0.473	3.691	<u>3.967</u>
		EDICT-Stage II	1.31	81.18	<u>0.933</u>	0.369	3.768	3.912
	Novelistic Description	Concat.	0.18	90.19	0.690	0.480	3.770	3.663
		Joint	<u>0.32</u>	54.51	0.936	0.489	3.736	4.171
		TED-TTS	0.73	75.46	<u>0.923</u>	<u>0.430</u>	3.715	<u>3.939</u>
		EDICT-Stage II	0.73	<u>87.67</u>	0.936	0.360	<u>3.750</u>	3.927
	Emotional Dialogue	Concat.	<u>1.10</u>	81.85	0.586	0.542	3.593	3.395
		Joint	0.84	61.48	0.827	<u>0.408</u>	3.622	<u>3.611</u>
		TED-TTS	2.04	<u>78.97</u>	0.799	0.418	3.656	3.559
		EDICT-Stage II	2.03	76.26	<u>0.808</u>	0.399	<u>3.632</u>	3.627
Chinese	Observational Narrative	Concat.	<u>2.26</u>	84.63	0.584	0.455	3.700	3.664
		Joint	1.87	62.41	0.874	<u>0.435</u>	3.744	3.825
		TED-TTS	3.24	75.93	<u>0.856</u>	0.462	3.711	<u>3.783</u>
		EDICT-Stage II	3.37	<u>79.96</u>	0.844	0.417	<u>3.726</u>	3.776
	Novelistic Description	Concat.	1.86	90.20	0.597	0.467	3.702	3.525
		Joint	<u>2.00</u>	60.55	<u>0.867</u>	<u>0.433</u>	<u>3.710</u>	3.853
		TED-TTS	2.76	74.06	0.874	0.466	3.727	3.752
		EDICT-Stage II	2.01	<u>83.66</u>	0.862	0.360	3.679	<u>3.760</u>